

Suppletion in Global Perspective

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This article documents and analyzes a system of suppletive alternations that are conditioned by top-down prosodic context. In Mandarin (Austronesian), seven heads supplete at the right edge of the phonological phrase to satisfy an output constraint on foot structure. When phrase-external phonological context makes it possible to resolve this output constraint in a more optimal way, these patterns of suppletion are suspended. These effects suggest that the mechanism that regulates suppletion, Vocabulary Insertion, must be situated within a phonological calculus that can access global context and respond to output constraints.

Keywords: suppletion, optimization, prosody, cyclicity, Sulawesi

1 Introduction

A central discovery of modern work in morphology is that morphosyntactic words are assembled within the morphosyntactic component of a modular, feed-forward grammar (Sadock 1980, Baker 1985, Hale and Keyser 1993, Halle and Marantz 1993, Marantz 1997). In the framework of Distributed Morphology (DM; Halle and Marantz 1993), a central component of this process is the operation that pairs morphosyntactic constituents with exponents: Vocabulary Insertion (VI). Halle and Marantz 1993 and many later works in DM propose that VI operates in the morphological module, as its outputs are often influenced by processes that seem to operate in that component of the grammar. But a parallel literature observes that VI often seems to be guided by output-oriented constraints in the phonology (Mester 1994, Mascaró 1996, 2007, Booij 1998, Nevins 2011). This observation raises the following question: at what derivational stage does VI apply?

The goal of this article is to push forward our understanding of VI by documenting and analyzing a set of suppletive alternations in Mandarin, an Austronesian language of Indonesia. In

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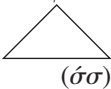
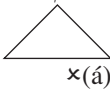
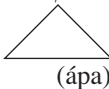
this language, there are seven X^0 s that take monosyllabic forms when they are followed by overt material in a small postsyntactic domain but supplete to disyllabic forms when they are not. This pattern is shown in the following examples with the root $\sqrt{\text{WHAT}}$: before the adverb *dua* ‘still’, this root takes the monosyllabic form *a* (1a), but when it appears alone before a predicate, it typically takes the disyllabic form *apa* (1b).

(1) *Top-down suppletion*

- a. {a .ua} {pilloa}?
 a dua pilloa
 what still ring
 ‘What’s still ringing?’
- b. {apa } {pilloa}?
 a pilloa
 what ring
 ‘What’s ringing?’

The investigation of this system will take us deep into the phrasal phonology of the language, as we search for a characterization of its domains and an explanation for its prosodic shape. We will ultimately see that these alternations are conditioned by surface phonological context as part of an output conspiracy: the phonological phrase (ϕ) requires its right edge to host a disyllabic foot in Mandar (2a), and this constraint bans short allomorphs like *a* (2b) and forces the insertion of long allomorphs like *apa* (2c) in this position.

(2) *The output-optimizing account*

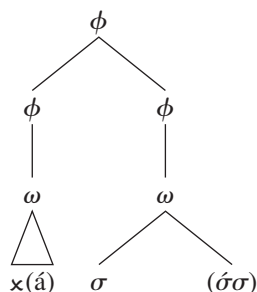
- a. ϕ

 ($\sigma\sigma$)
- b. ϕ

 ($\times(\acute{a})$)
- c. ϕ

 ($\acute{a}pa$)

As these patterns of suppletion are sensitive to the distribution of phonological phrases, I argue that VI must be able to access phonological information that sits beyond the c-command domains of the X^0 s that it targets—applying to at least some heads within the global phonological calculus where high-level prosodic structure is built. As these patterns of suppletion fit into an output conspiracy, in turn, I argue that VI must operate within a phonological calculus where it can be guided by output-oriented constraints (Mester 1994, Kager 1996, Booij 1998).

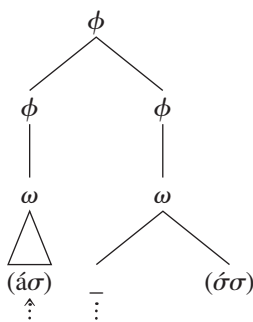
To strengthen this argument, I then demonstrate that the suppling heads in this system show the property termed *allomorphic vacillation* in Embick 2010: they retain their unmarked forms when their local context demands suppletion but the global context allows a more optimal repair. When the monosyllabic X^0 s in this system appear at the right edge of the ϕ , they take their unmarked monosyllabic allomorphs when it is possible to build a disyllabic trochee by pulling in an otherwise-unfooted syllable from the following ϕ . The result is the pattern of phonological displacement in tree (3).

(3) *Positional minimality: Optimally resolved by movement*

a. Larger context



b. Optimal repair



These results lead to two theoretical conclusions at the interface of morphology and phonology. First, VI must be rehoused in the phonology, where it can interact with output-oriented constraints as it selects the vocabulary entries to insert into particular X^0 s. Second, VI must be allowed to target particular X^0 s in a phonological evaluation that is global in character, where allomorph selection can make reference to nonlocal phonological information about the high-level prosodic organization of the clause.

The remainder of this article is structured as follows. Section 2 provides background information on Mandar and lays out the output conspiracy behind these effects. Section 3 introduces the alternations of interest, argues that they involve suppletion, and lays out an analysis that places VI in the phonology. Section 4 addresses the amount of the phonological context available to VI, developing a case for allomorphic vacillation that implicates a type of phonological movement on the way. Section 5 concludes.

2 Mandar and the Top-Down Constraint

Mandar is an Austronesian language of the South Sulawesi subfamily, spoken by 400,000 people in West Sulawesi, Indonesia (Grimes and Grimes 1987). The language shows a basic word order of VSO, derived by X^0 -movement of the verb to voice⁰ and covert movement of the absolutive subject to its high clause-internal A-position (Brodkin to appear). Mandar allows pro-drop and shows no morphological case marking. Transitive verbs host ergative agreement with the external argument, and every finite clause contains a second-position clitic that spells out agreement with the absolutive argument. The verb also shows alternations in transitivity that yield a type of Western Austronesian *voice system* (Aldridge 2004). A clause of the language is presented in example (4), showing the VSO word order and the system of agreement.¹

¹ Glossing: ABS: absolutive, ACC: accusative, AF: agent focus, ANTIP: antipassive, APPL: applicative, ERG: ergative, FUT: future, GEN: genitive, IPFV: imperfective, PFV: perfective. <c, ' > = /tʃ, ʔ/.

(4) *The shape of a Mandar clause*

na-alli i [s iKaco'] [o itim bunga] [x dionging].
3ERG-buy 3ABS NAME that flower yesterday
'Kacho' bought those flowers yesterday.'

This article reports on generalizations that have been gathered over two years of targeted study with Jupri Talib, a native speaker of Mandar with whom I have worked continually since 2019. These generalizations reflect the pronunciation that is introspectively judged to be optimal when speakers are asked to pronounce fully planned clauses in broad-focus contexts; they thus abstract away from the phonological deviations that are routinely forced by disfluencies in planning and production in naturalistic speech.² The data below have also been cross-checked several times in the field, presented to two audiences of native-speaker linguists (August 2023), and given as a talk to 150 Mandar-speaking undergraduates at the University of West Sulawesi (November 2023). The following discussion will focus on the prestige variety of Mandar, but the facts below seem to be uncontroversial in many Mandar dialects and are likely replicable in closely related languages in the South Sulawesi subfamily.³

2.1 *The Phonology of Stranding*

Our investigation begins with the phonology of stranding. In Mandar, interrogative *wh*-words and certain types of foci must move to the left periphery (Brodkin 2020). Example (5) shows this pattern and introduces the four-line format that will be used through this article, in which the top line shows the surface forms of constituents and the second line shows their underlying forms. This top line will be important as we study the conspiracy of Positional Minimality in section 2, various patterns of suppletion in section 3, and two further processes in the phrasal phonology in section 4.⁴

² The judgments in this study were collected as follows: the consultant was (1) presented with written examples in minimal pairs, (2) asked to reflect on their pronunciation at a regular speech rate with broad-focus prosody, and then (3) asked to produce each variant repeatedly until it was possible to judge each as *bisa dengan nada biasa* 'possible at a regular speech rate', *hanya dengan penekanan khusus* 'only possible with narrow focus', or *salah* 'incorrect'. The initial patterns were first noted in 2021, the larger system worked out in 2022, and the core generalizations cross-checked with different items several times in 2023.

³ Mandar has been the subject of much work in Indonesian by the Language Office of South Sulawesi (Sikki et al. 1987, Muthalib and Sangi 1991, Friberg and Jerniati 2000); on its phonology, see Jerniati 2005 and Pelenkahu, Muthalib, and Sangi 1983. The languages of the South Sulawesi subfamily have been described in a larger body of English-language work (B. Friberg 1991, 1996, Matti 1994, Strømme 1994, K. Valkama 1995, S. Valkama 1995, Finer 1997, 1998, 1999, Béjar 1999, Jukes 2006, Kaufman 2008, Lee 2008, D. Laskowske 2016). In Mamasa, an immediate neighbor of Mandar, Matti (1994) describes a positional minimality constraint that seems similar to the Mandar restriction described below (and see also Campbell 1989:19,160 on Pitu Ulunna Salu). Similar effects seem to exist in many other languages of the Northern Subgroup of the South Sulawesi subfamily, including at least some varieties of Mamuju, Ulumanda', Tae', and Enrekang; they are absent from at least certain varieties of Bugis and Makassarrese.

⁴ The relevant processes are described in detail in Brodkin to appear. They are (a) Coalescence: /ai ae/ → [e], /ao au/ → [o], (b) Gliding: prevocalic /i u/ → [j w], (c) Lenition: intervocalic /b d ɖ g/ → [w ɹ j ɣ], (d) Glottal coda deletion: intervocalic /ʔ/ → [Ø], intervocalic coda /ŋ/ → [Ø] + vowel nasalization, and (e) Nasal assimilation (Pater 1999): coda /ŋ/ → [C] before all nonnasal consonants but /b d ɖ g/.

(5) *Wh-movement in Mandar*

↓
pirap pe mjakke? ____?
piran pai miakke?
 when IPFV.3ABS leave
 ‘When did she leave?’

(B. Friberg and Jerniati 2000:178)

Wh-movement can target the complements of directional prepositions in Mandar. When this process strands monosyllabic P⁰s, these P⁰s are typically forced to expand. The following examples illustrate with the P⁰ *suŋ* ‘out’. Before overt complements, this P⁰ takes the form [suŋ] (6a). But when stranded, it undergoes V?-epenthesis to take the form [su?uŋ] (6b) (on V?-epenthesis elsewhere in South Sulawesi, see Clements 1986, Aronoff et al. 1987, McCarthy and Prince 1994, 1995, Broselow 1999, Gafos and Lombardi 1999, Kitto and de Lacy 1999, de Lacy 2006, Kawahara 2007, Stanton and Zukoff 2018).

(6) *P-stranding → Epenthesis*

- a. bemme i [PP **sun** [PP di pepattoan]].
 bemme i **suŋ** di pepattoan
 fall 3ABS out of window
 ‘It fell out of the window.’
- b. a nawemmei [PP **su?uŋ** ____]?
 a na-bemme-i **suŋ**
 what 3ERG-fall-APPL out
 ‘What did it fall out of?’

The examples in (7) introduce a second pattern of expansion. Mandar has many directional P⁰s whose unmarked forms are monosyllabic and contain the vowels /e o/. When these P⁰s are stranded, they retain underlying vowel sequences that are otherwise reduced by a regular process of vowel coalescence. For instance, the P⁰ ‘to’ in Mandar has the underlying form /lao/. Before overt complements, this P⁰ shows coalescence and surfaces as [lo] (7a). When it is stranded, however, it retains the form [lao] (7b). Similar alternations emerge with the P⁰s *sau* ‘to (oceanward)’ (which otherwise reduces to [so]), *nauŋ* ‘down to’ (otherwise [noŋ]), and *dai?* ‘up to’ (otherwise [de?]).

(7) *P-stranding → No coalescence*

- a. leppa ǎ? [PP **lo** [PP ɿi wojanna]].
 leppan a? lao di boɕaŋ-na
 go.over 1ABS to in house-3GEN
 ‘I went over to his house.’
 (Pelenkahu, Muthalib, and Sangi 1983:191)

- b. ne mukiriŋal [PP **lao** ____]?
 nai mu-kiriŋ-aŋ lao
 who 2ERG-send-APPL to
 ‘Who did you send it to?’

This initial effect is similar to an alternation that emerges in English with the P^0 s *to* and *for*. These P^0 s take subsyllabic forms when they appear before overt complements under default prosodic conditions, when the influence of focus and speech rate are properly controlled (8a). But when these P^0 s are stranded by *wh*-movement, they are forced to expand to prosodically larger forms (8b) (Selkirk 1995b).

(8) *P-stranding in English*

- a. I gave the books **tʰ/*to** Harvey.
 b. Who’d you give the books ***tʰ/to**?

These starting notes on *wh*-movement, in turn, form part of a larger generalization on the phonology of stranding. Mandarin also allows directional P^0 s to be stranded by a process of DP-ellipsis, and when this process targets the complements of these P^0 s, we see the same response: stranded P^0 s are typically forced to be disyllabic. The examples in (9) show that the P^0 *suŋ* ‘out’ undergoes the same type of Vʔ-epenthesis; other P^0 s are forced to retain underlying vowel sequences in the same context.

(9) *Ellipsis → Same effects*

- a. bemme i [PP **sun** [PP di tasmu ____]].
 bemme i suŋ di tas-mu
 fall 3ABS out of bag-2GEN
 ‘It fell out of your bag.’
 b. bemme i [PP **suʔuŋ** ____].
 bemme i suŋ
 fall 3ABS out
 ‘It fell out.’

We thus begin our investigation from the following generalization: in Mandarin, many P^0 s take monosyllabic forms before overt complements (10a) but become disyllabic when stranded (10b). I will refer to this restriction as the *Positional Minimality Constraint*.

(10) *The Positional Minimality Constraint*

- a. Overt complements b. Null complements
- ```

 PP
 / \
 sigma_P DP

```

```

 PP
 / \
 x sigma_P ____

```

## 2.2 The Right-Edge Effect

The Positional Minimality Constraint drives an identical set of alternations at the right edge of the Mandar DP. Mandar has many nominal roots that have the underlying shape CVV(C), where the VV sequence is one of /ai ae ao au/. When roots of this shape appear in the middle of the DP—for instance, when followed by adjectives—they show coalescence and take monosyllabic forms. But when they are final in the DP, they typically retain their underlying vowel sequences. The following examples illustrate: the N<sup>0</sup> *bau* ‘fish’ surfaces as *bo* before an overt A<sup>0</sup> (11a) but takes the form *bau* when final in the DP (11b).

### (11) Coalescence in the DP

- a. nande i **wo** pjapi.  
 na-ande i bau piapi  
 3ERG-eat 3ABS fish curried  
 ‘They ate the curried fish.’
- b. nande i **wau**.  
 na-ande i bau  
 3ERG-eat 3ABS fish  
 ‘They ate the fish.’

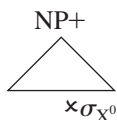
These alternations are matched in a second class of roots with the underlying shape CVV(C) in which the first vowel is one of [i u]. When roots of this second shape appear in the middle of the DP, they reduce to monosyllabic forms through a process that reduces prevocalic [i u] to [j w]. When they are final in the DP, however, they typically retain these underlying vowel sequences. The following examples illustrate: the N<sup>0</sup> *sia* ‘salt’ surfaces as *sya* before an overt A<sup>0</sup> (12a) but takes the form *sia* when final in the DP (12b).

### (12) Gliding in the DP

- a. nanne i **sja** mea.  
 na-anna-i i sia mea  
 3ERG-add-APPL 3ABS salt pink  
 ‘They added pink salt.’
- b. nanne i **sia**.  
 na-anna-i i sia  
 3ERG-add-APPL 3ABS salt  
 ‘They added salt.’

These restrictions suggest that the grammar of Mandar contains an output constraint that bans monosyllabic X<sup>0</sup>s at the right edge of a domain that corresponds roughly to the extended NP (Grimshaw 1991): P<sup>0</sup>s and N<sup>0</sup>s must be at least disyllabic in this position. This revised version of the Positional Minimality Constraint is sketched in diagram (13).

(13) *The Positional Minimality Constraint: Interim reformulation*



2.3 *The Top-Down Constraint*

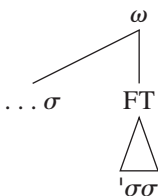
There is a straightforward way to understand the Positional Minimality Constraint in the phonology. Mandarin shows regular word-level penultimate stress, marked in the phonology with a low tone and with lengthening of the stressed syllable when it is metrically light (Pelenkahu, Muthalib, and Sangi 1983). Setting certain prefixes aside, every word hosts one instance of stress. The following examples illustrate this initial fact:

(14) *The distribution of word-level stress*

|                       |                 |                      |
|-----------------------|-----------------|----------------------|
| óro                   | oróaŋ           | oroáŋna              |
| oro                   | oro-aŋ          | oro-aŋ-na            |
| sit                   | sit-NOMINALIZER | sit-NOMINALIZER-3GEN |
| 'sit, seat, her seat' |                 |                      |

I thus propose that a disyllabic trochee is preferentially built at the right edge of every prosodic word ( $\omega$ ).<sup>5</sup> I assume that the syllables before the penult are generally not parsed into feet.<sup>6</sup> The result is the analysis of footing in (15).

(15) *The disyllabic trochee*



<sup>5</sup> Similar systems are described in the other languages of the South Sulawesi subfamily: regular word-level penultimate stress (Mithun and Basri 1986, Aronoff et al. 1987, Campbell 1989, T. Friberg and Friberg 1991, McCarthy and Prince 1994, 1995, Basri, Broselow, and Finer 1999, 2012, Broselow 1999, Jukes 2006, T. Laskowske 2007) and no iterative secondary stress (Campbell 1989, T. Friberg and Friberg 1991, Jukes 2006). The analysis in (15) thus follows an emerging consensus in work on other languages of the subfamily (Mithun and Basri 1986, T. Friberg and Friberg 1991, Basri, Broselow, and Finer 1999, 2012, Broselow 1999, Jukes 2006). Mandarin has no native nouns, verbs, or adjectives that synchronically retain the shape (C)V(C); it differs in this way from Makassarese (Jukes 2006).

<sup>6</sup> In principle, it is likely possible to derive the Positional Minimality Constraint without reference to feet by positing that the low tone linked to word stress must fall on a different syllable from a final H tone that falls at the right edge of the phonological phrase (example (22)). This account could derive the facts below by forcing the final word-level stress in the phonological phrase to fall one syllable before the right edge. Nevertheless, the foot in (15) is independently attested in the prosodic morphology of Mandarin. There is a pattern of reduplication that copies a two-syllable sequence from the left edge of the base and superimposes trochaic stress on that string (/balao/ 'rat' → [bála-baláo]), as if to fill out the same disyllabic trochee. I take this pattern to show that this foot exists in Mandarin and thus employ it in the analysis below.



This disyllabic trochee is forced into existence through the interaction of two pressures. The first is the constraint HEADEDNESS (Nespor and Vogel 1986), which demands that prosodic constituents contain heads (16a)—ruling out  $\omega$ s that do not contain feet. The second is the constraint FOOTFORM (16b), which demands that metrical feet take the form of the disyllabic trochee in (15).

(16) *Forcing the construction of disyllabic trochees*

a. HEADEDNESS

Assign one violation for every nonterminal output prosodic category that does not immediately dominate a prosodic category of equal or immediately lesser size on the prosodic hierarchy  $\iota > \phi > \omega > \text{FOOT} > \sigma > \mu$ .

b. FOOTFORM

Assign one violation for every foot that is not a disyllabic trochee.

From this perspective, we can understand the Positional Minimality Constraint as a more stringent version of the constraint FOOTFORM. In the terminology of Optimality Theory (OT; Prince and Smolensky 1993), it is clear that FOOTFORM must be a ranked and violable constraint: the language allows monosyllabic feet in certain positions, as CVV(C) nouns continue to bear stress but reduce to monosyllabic forms in specific positions. We can derive this result by ranking the general version of FOOTFORM beneath the constraint that drives coalescence and gliding, which I take to be ONSET (17a) (Ito 1986, Rosenthall 1994; cf. Borroff 2007). This interaction is shown in tableau (17b).

(17) *Deriving monosyllabic feet*

a. ONSET

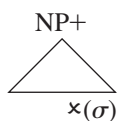
Assign one violation for every syllable that lacks an onset.

b.

| sia mea ‘pink salt’                       | ONSET | FOOTFORM |
|-------------------------------------------|-------|----------|
| ☞ a. [ $\omega$ (sǰá)] [ $\omega$ (mé.a)] |       | *        |
| b. [ $\omega$ (sí.a)] [ $\omega$ (mé.a)]  | *!    |          |

At the right edge of the extended NP, the constraint FOOTFORM must be enforced in a stricter way: the final word must be disyllabic because the final foot must be a disyllabic trochee. The result is an analysis that allows us to understand the Positional Minimality Constraint as a word minimality effect that emerges from constraints on the structure of feet (Ito and Mester 1992, McCarthy and Prince 1993). The following diagram illustrates this analysis, using parentheses to mark the boundaries of feet:

(18) *The Positional Minimality Constraint: Second reformulation*



This revision allows us to make quick sense of the patterns that we have seen so far. Coalescence and gliding are blocked at the right edge of the extended NP by the ranking of ONSET beneath this more stringent form of FOOTFORM, which is restated in (19a). Tableau (19b) shows how this ranking prevents CVV(C) nouns from reducing to monosyllables at the right edge of the relevant domain.

(19) *DP-internal restrictions: The starting analysis*

a. POSITIONAL MINIMALITY (POS-MIN)

Assign one violation for every foot at the right edge of the extended NP that is not the disyllabic trochee in (15).

|    |                                       |       |         |
|----|---------------------------------------|-------|---------|
| b. | nande i [DP bau] ‘They ate the fish.’ | ONSET | POS-MIN |
|    | ☞ a. nande i {ba.u}                   |       | *       |
|    | b. nande i {bo}                       | *!    |         |

This reanalysis also allows us to understand the precise distribution of the Positional Minimality Constraint. The positions that enforce the Positional Minimality Constraint in Mandarin have a consistent prosodic signature: they invariably host a high tone (H). The examples in (20) illustrate: the extended NP always hosts exactly one H tone of this type, falling after nouns when they are final (20a), falling after the rightmost adjective when adjectives are present (20b), and persisting in the presence of higher P<sup>0</sup>s (20c).

(20) *The phrase-final H tone*

- a. málli<sup>H</sup> i wáu<sup>H</sup>.  
maŋ-alli i bau  
ANTIP-buy 3ABS fish  
‘She bought fish.’
- b. málli<sup>H</sup> i wó pjápi<sup>H</sup>.  
maŋ-alli i bau piapi  
ANTIP-buy 3ABS fish curried  
‘She bought curried fish.’
- c. bémme<sup>H</sup> i sun di pepattóan<sup>H</sup>.  
bemme i suŋ di pepattoan  
fall 3ABS out of window  
‘It fell out of the window.’

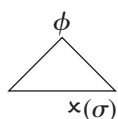
The same H tone falls at the right edges of stranded P<sup>0</sup>s, as well.

(21) *Phrase-final H tone: After stranded P<sup>0</sup>s*

- naláttfar<sup>H</sup> i sú?uŋ<sup>H</sup> itis sanéke kettfu<sup>?</sup> bukú?u<sup>H</sup>.  
na-lattfar i suŋ itij sanaeke kettfu? buku-?u  
3ERG-throw 3ABS out that kid small book-1GEN  
‘That little kid threw out my book.’

This correlation suggests that the H tone and the Positional Minimality Constraint are keyed to the right edge of a single phonological domain: an abstract phonological constituent on the prosodic hierarchy (Selkirk 1978, 1984, 1986, Nespor and Vogel 1986, Ito and Mester 2007, 2012, 2013) that is roughly isomorphic to the extended NP. There is a natural candidate for such a constituent, and it is the phonological phrase ( $\phi$ ). This identification allows us to understand the Positional Minimality Constraint as a typical requirement of positional binarity, forcing a lower prosodic constituent to branch when it appears at the edge of a higher prosodic domain (Downing 1998, Booij 1998, de Lacy 2001, Parker 2001, Smith 2002, 2004, Prieto 2005, Elordieta 2006). On this analysis, the Positional Minimality Constraint can be understood as a requirement that the  $\phi$  host a final trochee that contains two  $\sigma$ s (22).

(22) *The Positional Minimality Constraint: Final reformulation*



The  $\phi$ -level analysis also allows us to understand why the Positional Minimality Constraint is enforced at the right edges of DPs and stranded P<sup>0</sup>s. Phonological phrases are built to satisfy an interface requirement to preserve syntactic XPs (Selkirk 1984, 1986, 1995a,b, 2009, 2011, Nespor and Vogel 1986, Selkirk and Tateishi 1988, 1991, Selkirk and Shen 1990). Following Ito and Mester (2019), I will assume that this requirement is one of existential correspondence: every input XP must be placed in correspondence with an output  $\phi$  (23a). These  $\phi$ s are aligned with their corresponding XPs by constraints of the family PROSODY-TO-SYNTAX MATCH (Selkirk 2011, Elfner 2015), like MATCH( $\phi$ ,XP) (23b).

(23) *The phrase-level mapping constraints*

a. MAX-XP

Assign one violation for every XP in an input syntactic representation S that does not correspond to a  $\phi$  in a corresponding output phonological representation P.

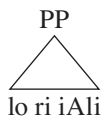
b. MATCH( $\phi$ ,XP)

Assign one violation for every  $\phi$  in an output phonological representation P whose edges are not aligned with the edges of a corresponding XP in an input syntactic representation S.

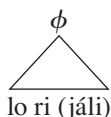
The constraint MAX-XP demands that all DPs and PPs map to  $\phi$ s. MATCH( $\phi$ ,XP) then demands that the edges of these  $\phi$ s align with the edges of their corresponding XPs. When not dominated by higher-ranking phonological constraints, MAX-XP and MATCH( $\phi$ ,XP) will then force the right edges of DPs and PPs to fall at the right edges of  $\phi$ s, as sketched in the pair of trees in (24). The PP *lo ri iAli* ‘to Ali’ in tree (24a), for instance, should be mapped to the isomorphic  $\phi$  in (24b).

(24) *Mapping PPs to  $\phi$ s*

a. Syntax



b. Prosody



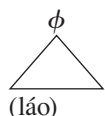
The same mapping of  $PP \rightarrow \phi$  persists in contexts of *wh*-movement and ellipsis. MAX-XP mandates that all XPs—including functional projections like PPs—be placed in correspondence with  $\phi$ s (Ito and Mester 2019; see also Elfner 2015). As a result, PPs should be mapped to  $\phi$ s in Mandarin even when  $P^0$ s are stranded by complement ellipsis. In this context, stranded  $P^0$ s should surface at the right edges of  $\phi$ s that correspond to their PPs (25b) and should thus be subjected to the Positional Minimality Constraint.

(25) *Stranded  $P^0$ s  $\rightarrow \phi$ s*

a. Syntax



b. Prosody



The tableau in (26) illustrates the derivation of the particular phonology that emerges in this context. The Positional Minimality Constraint is defined in OT terms in (26a). For this constraint to force V?-epenthesis in the stranded  $P^0$  *suŋ*, it must be ranked above the constraint DEP (26b). The constraint MAX-XP forces the stranded  $P^0$  to form a  $\phi$ , and the constraint HEADEDNESS (16a) demands that this  $\phi$  contain a  $\omega$ . The combined effect of these constraints is to force the selection of a candidate in which the stranded  $P^0$  hosts a disyllabic foot and forms an independent  $\omega$  and  $\phi$ .

(26) *Deriving V?-epenthesis*

a. POSITIONAL MINIMALITY (POS-MIN; formally ALIGN-R ( $\phi, (\acute{\sigma}\sigma)$ ))

Assign one violation for every  $\phi$  that is not right-aligned with a disyllabic trochee.

b. DEP

Assign one violation for every output segment that lacks an input correspondent.

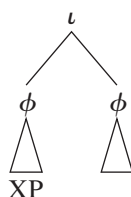
| c.   | bemme i suŋ ‘It fell out.’                                                                           | MAX-XP | POS-MIN | HEAD | DEP |
|------|------------------------------------------------------------------------------------------------------|--------|---------|------|-----|
| ☞ a. | $\{\phi[\omega(\acute{b}\acute{e}mme)]\} \text{ i } \{\phi[\omega(\acute{s}\acute{u}?\acute{u}ŋ)]\}$ |        |         |      | **  |
| b.   | $\{\phi[\omega(\acute{b}\acute{e}mme)]\} \text{ i } \{\phi(\acute{s}\acute{u}?\acute{u}ŋ)\}$         |        |         | *!   | **  |
| c.   | $\{\phi[\omega(\acute{b}\acute{e}mme)]\} \text{ i } \{\phi \text{ suŋ}\}$                            |        | *!      |      |     |
| d.   | $\{\phi[\omega(\acute{b}\acute{e}mme)]\} \text{ i suŋ}$                                              | *!     |         |      |     |

## 2.4 Surface Orientation

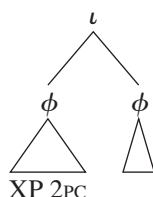
The Positional Minimality Constraint must be enforced in an output-oriented phonological calculus. This is because it applies exactly in the positions that fall at the right edges of  $\phi$ s in the surface phonology—and not, therefore, at the right edges of all syntactic XPs. We can see this effect by studying cases where the boundaries of  $\phi$ s are forced to diverge from the boundaries of their corresponding XPs by requirements of prosodic subcategorization (Inkelas 1989). For instance, Mandarin has a set of second-position elements that form minimal  $\phi$ s with their hosts. These elements thus create a systematic syntax-prosody mismatch: when introduced into clauses where the initial  $\phi$  should correspond neatly to an XP (27a), the initial  $\phi$  becomes misaligned (27b).

### (27) Second-position constituents (2PC) $\rightarrow$ Mismatches

a. Default prosody



b. Mismatch prosody



Example (28) illustrates this behavior with the second-position element *nasay* ‘all, every’. In finite clauses, this element shifts to second position and forms a minimal  $\phi$  with the constituent that precedes it (as shown by the distribution of the H tone). In this example, *nasay* floats out of a subject to form a minimal  $\phi$  with the initial V.

### (28) Second-position mismatches

|          |                          |          |       |    |                    |     |           |                       |        |
|----------|--------------------------|----------|-------|----|--------------------|-----|-----------|-----------------------|--------|
| míssun   | <b>násam<sup>H</sup></b> | mi       | dome  | .i | kámar <sup>H</sup> | tó  | nasoláŋam | maraʔdíá <sup>H</sup> | _____. |
| missuŋ   | <b>nasay</b>             | mi       | domai | di | kamar              | tau | nasolanaŋ | maraʔdia              |        |
| come.out | every                    | PFV.3ABS | out   | of | room               | guy | with      | king                  |        |

‘Every one of the men with the king came out of the room.’  
 (Sikki et al. 1987:1083)

The construction of a misaligned  $\phi$  around *nasay* and its host must be forced by a lexical property of *nasay*: this  $X^0$  must be parsed into a minimal  $\phi$  with the element to its left (29a). I take this requirement to be enforced by the output constraint SUBCAT (29b), which forces the idiosyncratic phonological requirements of  $X^0$ s to be met in the surface phonology (Bonet, Lloret, and Mascaró 2007, Bennett, Harizanov, and Henderson 2018). Ranked above MATCH( $\phi$ ,XP), this constraint will force *nasay* to form a minimal  $\phi$  with the preceding constituent (29c).

### (29) Forcing mismatches with *nasay*

a. Subcategorization frame

$\sqrt{\text{EVERY}} \rightarrow \{\phi_{[\text{MIN}]} \dots \text{nasang}\}$

b. SUBCAT

Assign one violation for every  $X^0$  whose phonological subcategorization frame is not satisfied in the output phonological representation.

c.

| V nasaj . . . DP $t_{\text{nasaj}}$                                                         | SUBCAT | MATCH( $\phi, XP$ ) |
|---------------------------------------------------------------------------------------------|--------|---------------------|
| a. $\{\phi_{[\text{MIN}]}\text{ v nasaj}\}\{\phi\text{ DP}\}$                               |        | *                   |
| b. $\{\phi_{[\text{MIN}]}\text{ v}\}\{\phi_{[\text{MIN}]}\text{ nasaj}\}\{\phi\text{ DP}\}$ | !*     |                     |

We can now see the effect of this mismatch in the nominal system. When focused DPs move to the left periphery in Mandar, they continue to form  $\phi$ s and the Positional Minimality Constraint continues to apply at their right edges. As a result, roots of the shape CVV(C) are forced to remain disyllabic in this position (30a). But when these DPs are followed by *nasaj*, they reduce to monosyllabic forms (30b).

(30) *Second-position elements shield DPs*

- a. **sía<sup>H</sup>** kápan<sup>H</sup> naluppéi<sup>H</sup> \_\_\_\_.  
sia kapaj na-luppei \_\_\_\_  
salt maybe 3ERG-forget  
‘Maybe they forgot salt.’
- b. **sjá násak<sup>H</sup>** kápan<sup>H</sup> naluppéi<sup>H</sup> \_\_\_\_.  
sia nasaj kapaj na-luppei \_\_\_\_  
salt every maybe 3ERG-forget  
‘Maybe every one of them forgot salt.’

Within this misaligned clause-initial  $\phi$ , the Positional Minimality Constraint then takes effect at the right edge. The examples in (31) illustrate with the adverb *to* ‘also’, which also appears in second position and forms a minimal  $\phi$  with its host. This adverb stays monosyllabic when it is shielded from the right edge of the  $\phi$  by other second-position elements (31a). But when it appears at the right edge of the  $\phi$ , it typically responds to the Positional Minimality Constraint and shows V?-epenthesis (31b).

(31) *Second-position elements react to the Positional Minimality Constraint*

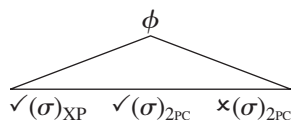
- a. jó **tó** wándi<sup>H</sup> mítǽǽ?<sup>H</sup>.  
jau to bandi miŋ-tǽǽ?  
I also certainly ANTIP-follow  
‘I’m certainly coming too.’
- b. jó **tóʔo<sup>H</sup>** mítǽǽ?<sup>H</sup>.  
jau to miŋ-tǽǽ?  
I also ANTIP-follow  
‘I’m coming too.’

These interactions reveal that the Positional Minimality Constraint operates transparently over the surface prosody. When DPs are focus-fronted and followed by second-position elements,

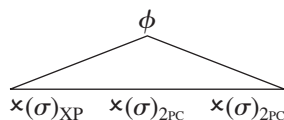
the Positional Minimality Constraint is exclusively enforced at the right edges of the surface  $\phi$ s. This distribution holds even when the relevant  $\phi$ s are not well-aligned with syntactic domains, as shown in the following diagram:

(32) *Positional Minimality Constraint*  $\rightarrow$  *Surface prosody*

a. Attested enforcement



b. Unattested enforcement



This conclusion suggests that the Positional Minimality Constraint must be evaluated at a derivational stage in which, at a minimum, DPs have been focus-fronted and second-position elements have been linearized and parsed into  $\phi$ s with their hosts. This result reveals that the Positional Minimality Constraint cannot be enforced cyclically over any putative DP-internal cycles of prosodic structure-building. We thus arrive at an architectural understanding that converges with the results of much phonological research on the cycle, which has found little to no evidence for the cyclic application of phrasal phonology (Kiparsky 1982, Bermúdez-Otero 2012) or the cyclic construction of  $\phi$ -level prosodic domains (Cheng and Downing 2016). Its force is to suggest, then, that the Positional Minimality Constraint must be evaluated exactly one time—most naturally in the global and surface-oriented phonological calculus in which prosodic structure is built (McCarthy and Prince 1993, Selkirk 1995a, 2009, 2011).

### 3 Top-Down Suppletion

Up to this point, we have seen that monosyllabic  $X^0$ s undergo a range of segmental repairs at the right edge of the  $\phi$  in Mandarin to satisfy a positional constraint on word minimality. The next stage of our investigation will turn to a second class of monosyllabic  $X^0$ s that respond to this constraint in a different way: suppletion.

#### 3.1 Idiosyncratic Alternations

Mandarin has a process of NP-ellipsis that can strand demonstratives, like the monosyllabic *ndi* ‘this’ (in hand), *de* ‘this’, and *do* ‘that’. The demonstratives that are stranded in this way are forced to form  $\phi$ s, much like the  $P^0$ s stranded by DP-ellipsis above. But the monosyllabic demonstratives in the language do not respond to the Positional Minimality Constraint by undergoing V?-epenthesis or retaining vowels that would otherwise be erased by productive processes in the synchronic phonology. Instead, they alternate in segmentally idiosyncratic ways. When the monosyllabic *Dem*<sup>0</sup> *ndi* is stranded, for instance, it takes the disyllabic form *indi* (33b).

(33) *Irregular expansion: Ndi*–*indi*

- a. mélo<sup>H</sup>      uwéŋan<sup>H</sup> [<sub>DP</sub> **ndi** wúku<sup>H</sup> é<sup>H</sup> ].  
 melo? o      u-beŋaŋ      ndi buku      e  
 want 2ABS 1ERG-give      this book      here  
 ‘I want to give you this book here.’

- b. mélo<sup>H</sup>      uwéŋa<sup>H</sup>    [DP **índi**<sup>H</sup> \_\_\_\_ é<sup>H</sup> ].  
 melo? o      u-beŋaŋ      ndi      \_\_\_\_ e  
 want 2ABS 1ERG-give      this      here  
 ‘I want to give this (thing) here to you.’

The demonstratives *de* and *do*, in turn, expand to the respective forms *di?e* and *di?o*.

(34) *Irregular expansion: Do–di?o*

- a. mókaʔ<sup>H</sup>      mála<sup>H</sup> i      [DP **ɬo** síttʃoʔ<sup>H</sup> ó<sup>H</sup> ].  
 moka aʔ    maŋ-ala i      do síttʃoʔ o  
 refuse 1ABS AF-take 3ACC      that speck there  
 ‘I refuse to take that microscopic amount there.’  
 (Sikki et al. 1987:664)
- b. sápa<sup>H</sup>      muwajára<sup>H</sup>    ĩ      [DP **ɬíʔo**<sup>H</sup> \_\_\_\_ ó<sup>H</sup> ]?  
 saŋapa    mu-baɟar-aŋ i      do      \_\_\_\_ o  
 how.much 2ERG-pay-APPL 3ACC      that      there  
 ‘How much did you pay for that (thing) there?’  
 (B. Friberg and Jerniati 2000:304)

An analogous alternation emerges under VP-ellipsis (VPE). In Brodtkin to appear, I show that Mandar allows ellipsis of the complements of aspectual auxiliaries and negators. This process also forces stranded auxiliaries and negators to form  $\phi$ s, and in this context, the Positional Minimality Constraint forces regular alternations in many monosyllabic  $X^0$ s of this type. For instance, the prohibitive *da* shows Vʔ-epenthesis (35).

(35) *VPE: Regular expansion of Pol<sup>0</sup>*

- a. **da**      lámba<sup>H</sup>!  
 da      lamba  
 PROHIBITIVE go  
 ‘Don’t go!’
- b. mála<sup>H</sup> o      lámba<sup>H</sup> téʔe<sup>H</sup>, tapi **ɬáʔa**<sup>H</sup>      \_\_\_\_ maróndonɿ.  
 mala o      lamba teʔe    tapi da      lámba marondonɿ  
 can 2ABS go      now    but PROHIBITIVE go      tomorrow  
 ‘You can go now, but don’t go tomorrow.’

But the realis negator *ndaŋ* shows a segmentally idiosyncratic alternation in this context: when stranded by this type of ellipsis, this  $X^0$  takes the form *andiaŋ*.

(36) *Irregular expansion: Ndaŋ–andiaŋ*

- a. do anɟóro<sup>H</sup> [POLP **ndaŋ** ŋúra wéya<sup>H</sup> ó<sup>H</sup> ].  
 do anɟoro      ndaŋ ŋura bega o  
 that coconut      not unripe too      there  
 ‘That coconut there is not too unripe.’



b. De anjoro ngura bega e, . . .

‘This coconut is too unripe, . . .

tapi ɔ andʒoro<sup>H</sup> [<sub>PolP</sub> **andían**<sup>H</sup> \_\_\_\_ \_\_\_\_ ó<sup>H</sup> ].

tapi do andʒoro ndaŋ ɲura bega o

but that coconut not unripe too there

‘. . . but that coconut there is not too-unripe.’

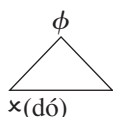
These segmentally irregular alternations have a familiar prosodic shape: they arise in  $X^0$ s that are typically monosyllabic and they render them disyllabic at the right edge of the  $\phi$ . In this respect, it seems natural to imagine that they operate in service of the Positional Minimality Constraint, which bans the unmarked short forms (37a) at the right edge of the  $\phi$  (37b) and forces the long forms to appear (37c).

(37) *Stranded Dem<sup>0</sup>/Neg<sup>0</sup> → Suppletion*

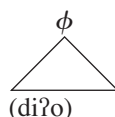
a. Default



b. Constraint



c. Repair



In their segmental irregularity, these alternations are matched by a parallel in English. The English  $P^0$  of typically takes the shape /ə/ before overt complements but expands irregularly to /ʌv/ when stranded (38): presumably because the PP must always form a  $\phi$  and that  $\phi$  must always contain a licit  $\omega$  (due to HEADEDNESS (16a); Ito and Mester 2019).

(38) *Irregular expansion: ə-ʌv*

a. Can I get a cup [ə/\*ʌv] coffee?

b. What do you want a cup [\*ə/ʌv] \_\_\_\_?

The same types of alternations can be seen in the English system of VPE (King 1970, Kaisse 1985, Selkirk 1995a, Anderson 2008, Ito and Mester 2019). Many English auxiliaries have unmarked forms that are subsyllabic, including the proclitic *sh* ‘<should>’, *c* ‘<could>’, *w* ‘<would>’, and *d* ‘<do/did>’ and the enclitic *’m* ‘<am>’, *’re* ‘<are>’, *’s* ‘<is/has>’, *’ve* ‘<have>’, *’ll* ‘<will>’, and *’d* ‘<would/had/did>’. These auxiliaries must take their unmarked forms when they appear in  $T^0$  or  $C^0$  and the influence of focus is properly controlled. But in contexts of VPE, these auxiliaries expand in segmentally unpredictable ways to prosodically larger forms (39): once again because the XPs that they head must form  $\phi$ s and must contain  $\omega$ s (Anderson 2008, Ito and Mester 2019).<sup>7</sup>

<sup>7</sup> The choice of allomorph for ‘<would>’ seems to depend on even finer-grained facts of prosodic phrasing: *-d* in the configuration  $\{\phi_{[MAX]} \text{ Subj Aux VP}\}$  and *w*- otherwise (e.g.,  $\{\phi_{[MAX]} \text{ Subj}\} \{\phi_{[MAX]} \text{ Aux VP}\}$ ). In English, the patterns of suppletion in this system seem to coexist with alternations that are more properly phonological, as many auxiliaries can take forms that fall between the two clear extremes (e.g., /hæv/-/həv/-/əv/-/v/) (Zwicky 1970, 1985). Patterns of this second type are likely best handled with lexically indexed constraints or more complex underlying representations. For arguments that the relevant alternations in Mandarin involve suppletion, see section 3.2.

(39) *King's Generalization: English auxiliaries expand before VPE sites*

- a. We'd definitely leave, and they [\*'d / wʌd] \_\_\_\_ too.
- b. You sh'definitely go, and we [\*sh' / shʌd] \_\_\_\_ too.
- c. They'll definitely try it, and we [\*'ll / wɪl] \_\_\_\_ too.

In the shapes of the Mandarin alternations, in turn, we can detect a second parallel in the Mandarin *wh*-system. In Mandarin, the root  $\sqrt{\text{WHAT}}$  takes the form *a* when it is nonfinal in the  $\phi$ —for instance, when followed by a  $\phi$ -internal second-position adverb like *dua* ‘still’ (40a). But when final in a  $\phi$ , it typically expands to the disyllabic *apa* (40b).

(40) *Irregular expansion: A–apa*

- a. **á**      júa<sup>H</sup> píllóa<sup>H</sup>?  
       a      dua    piŋ-loa  
       what still    ANTIP-SOUND  
       ‘What’s still ringing?’
- b. **ápa**<sup>H</sup> píllóa<sup>H</sup>?  
       a      piŋ-loa  
       what    ANTIP-SOUND  
       ‘What’s ringing?’

A similar alternation emerges in the root  $\sqrt{\text{WHICH}}$ . This root takes the monosyllabic form *na* when it is nonfinal in the  $\phi$ : for instance, when it is followed by *nasaj* (41a). But when stranded at the right edge of the  $\phi$ , it typically takes the form *inna* (41b).

(41) *Irregular expansion: Na–inna*

- a. **ná**      násam<sup>H</sup> mweŋéi<sup>H</sup>?  
       na      nasaj    mu-eŋei  
       which every    2ERG-be.in  
       ‘Which place are you guys all in?’
- b. **inna**<sup>H</sup> wémme<sup>H</sup>?  
       na      bemme  
       which fall  
       ‘Which one fell?’

A similar alternation can be seen in the word for *who*, which is at least historically built from  $\sqrt{\text{WHICH}}$  and a suffixed proprial article. In  $\phi$ -medial positions, ‘who’ takes the form *ne* (underlyingly /nai/) (42a). In  $\phi$ -final positions, however, ‘who’ does not respond to the Positional Minimality Constraint by retaining its underlying vowel sequence; instead, it takes the form *inne* (underlyingly /innai/) (42b).

(42) *Irregular expansion: Na–inna*

- a. **né**      násam<sup>H</sup> mwíta<sup>H</sup>?  
       nai    nasaj    mu-ita  
       who every    2ERG-see  
       ‘Who did every one of you guys see?’

b. **inne**<sup>H</sup> lámba<sup>H</sup>?

nai lamba

who go

‘Who left?’

Within the second-position system, finally, we can note one last pattern of a clearly related shape. Mandarin has an adverb *a* ‘maybe’ that surfaces within the first  $\phi$  in the clause. This adverb can appear whenever it is not absolutely final in the  $\phi$ : for instance, when it is followed by other second-position elements in this position (43a). But when it should appear at the right edge of a  $\phi$ —for instance, when no other second-position elements follow—its appearance is banned (43b). The result is an apparent case of phonological blocking (Raffelsiefen 1996, 1999): this  $X^0$  cannot respond to the Positional Minimality Constraint on its own, so whenever it should be final in a  $\phi$ , it is suppressed.

(43) *Phonological blocking: A ‘maybe’*

a. ikáʃo á ɿa<sup>H</sup>mo máttímbe<sup>H</sup>.

ikaʃo? a damo maŋ-timbe

NAME maybe just ANTIP-throw

‘Maybe it was Kacho’ who just threw something.’

(Sikki et al. 1987:439)

b. \*ikáʃo á(?a)<sup>H</sup> máttímbe<sup>H</sup>.

ikaʃo? a maŋ-timbe

NAME maybe ANTIP-throw

Intended: ‘Maybe it was Kacho’ who threw something.’

### 3.2 Top-Down Suppletion

The seven alternations across these systems in Mandarin are summarized in table (44). These alternations have a consistent prosodic shape: each operates at the right edge of the  $\phi$  and each eliminates monosyllabic feet in that position. In the realm of segmental phonology, however, they are united only by a thin surface generalization: the unmarked member of each pair is built from a subset of the segmental content of its longer form.

(44) *Suppletive monosyllables*

| Gloss          | this (in hand) | this | that | NEG    | what | which | who   |
|----------------|----------------|------|------|--------|------|-------|-------|
| $\phi$ -medial | ndi            | de   | do   | ndaŋ   | a    | na    | nai   |
| $\phi$ -final  | indi           | diʔe | diʔo | andiaŋ | apa  | inna  | innai |

I propose that these alternations are suppletive in the synchronic grammar of Mandarin. On this view, the short forms of these  $X^0$ s are not derived from their longer counterparts by synchronic reduction. Rather, each pair contains two vocabulary entries—an unmarked allomorph and a marked alternative—that are selected in different contexts as the phonology demands. As these patterns of suppletion are conditioned by top-down prosodic context, I will refer to them as cases of *top-down suppletion*.

To secure the identification of these alternations as suppletive, we can weigh this analysis against two alternative approaches to alternations of this type. The first of these would take the relevant alternations to reflect item-specific rules of readjustment (Halle and Marantz 1993): plausibly, rules that applied at a derivational stage that followed VI and reduced long allomorphs whenever they were nonfinal in the  $\phi$ . Examples of such rules are given in (45), focusing on the alternations *de–di?e* and *a–apa*.

(45) *Rejected: The Readjustment Rule analysis*

- a. Readjustment of *di?o*:  $i? \rightarrow \emptyset / \{\phi \text{ — } \dots\} \dots [\sqrt{\text{THAT}}]$
- b. Readjustment of *apa*:  $p \rightarrow \emptyset / \{\phi \text{ — } \dots\} \dots [\sqrt{\text{WHAT}}]$

Four widespread considerations suggest that there is no room in a predictive theory of morphology for readjustment rules like those of (45). First, there is no way to draw a principled synchronic distinction between the alternations that would involve readjustment and the alternations that would involve suppletion (Haugen and Siddiqi 2013, 2016, Haugen 2016; see also Mel'čuk 1994, Corbett 2007). Second, there is no need to appeal to readjustment rules to capture morphophonological generalizations that hold over subsets of lexical items (cf. Harley and Tubino Blanco 2013), as such patterns can be handled with theoretical mechanisms that are independently necessary in phonology, such as lexical stratification (Ito and Mester 1995) and complex underlying representations (Bermúdez-Otero 2012, Bye and Svenonius 2012). Third, readjustment rules blur the lines between morphology and phonology in a manner that makes it difficult to investigate fine interactions between the two. Fourth, readjustment rules are unconstrained by definition and thus substantially undercut the predictive capacity of the theories that employ them (Bermúdez-Otero 2012).

Stepping past the readjustment rule analysis, we might then consider a second possibility: to treat the alternations in (44) as the workings of the synchronic phonology. On such an account, these alternations would involve patterns of reduction that applied within the phonology and acquired their particular shapes from the interaction of general phonological constraints with the particular underlying representations of the long allomorphs in this system. The specific segmental processes that would have to operate in the synchronic phonology, on such an analysis, are given in (46).

(46) *Rejected: Synchronic phonological analysis*

- a. Vnd-aphaeresis: */andian/*  $\rightarrow$  *[ndaŋ]*, */indi/*  $\rightarrow$  *[ndi]*
- b. Vnn-aphaeresis: */inna/*  $\rightarrow$  *[na]*, */innai/*  $\rightarrow$  *[nai]*
- c. diV-syncope: */di?e/*  $\rightarrow$  *[de]*, */di?o/*  $\rightarrow$  *[do]*, */andian/*  $\rightarrow$  *[ndaŋ]*
- d. P-deletion: */apa/*  $\rightarrow$  *[a]*

There are many ways to derive phonological alternations with the shapes in (46), but there is no independent evidence internal to Mandar to suggest that these phonological alternations synchronically exist. First, there are no other lexical items that show alternations of these shapes. Focusing first on aphaeresis, we can observe that Mandar has many disyllabic roots that begin in the sequences */ind/* and */and/*. Across categories, these roots can never reduce to forms that

begin with the sequence [nd]. The examples in (47) illustrate: even in the middle of the  $\phi$ , high-frequency nouns like *indo?* ‘mother’ and *ande* ‘food’ cannot show any type of Vnd-aphaeresis.

(47) *No Vnd-aphaeresis*

- a. malólo sánnaʔ<sup>H</sup> i \*(i)ndoʔ rannúaŋ<sup>H</sup>.  
 malolo sannal i indoʔ rannuaŋ  
 beautiful very 3ABS mother yearning  
 ‘The Yearning Mother (a local waterfall) is really beautiful.’
- b. á na<sup>H</sup> máppapía<sup>H</sup> o \*(á)nde maráras<sup>H</sup>?  
 a na maŋ-papia o ande mararas  
 what and ANTIP-make 2ABS food spicy  
 ‘Why are you making spicy food?’

There is also no general process of Vnn-aphaeresis in the language. Disyllabic roots that begin in the sequence /Vnn/ cannot take forms that begin with the sequence [(n)n]. Even  $\phi$ -medially, for instance, the V *amme?* ‘swallow’ cannot lose its initial vowel.

(48) *No Vnn-aphaeresis*

- \*(ám)meʔ násaŋ<sup>H</sup> i itip paúlli<sup>H</sup>.  
 ammeʔ nasaf i itif paulli  
 swallow every 3ABS that medicine  
 ‘Take every one of those medicines.’

Turning now to syncope, we can observe that there is no other context in Mandar in which the sequence /di(?)V/ is reduced to [dV]. Every lexical root that contains the sequence /diV/ requires hiatus to be resolved by gliding /i/ to [j], rather than by deleting /i/.

(49) *No diV deletion*

- d\*(j)óm bándi<sup>H</sup> di lémbon<sup>H</sup> d\*(j)óniŋ<sup>H</sup>?  
 dioŋ bandi di lembon dionif  
 down certainly.3ABS in river yesterday  
 ‘Was she really down by the river yesterday?’

Finally, there is no process in the language that deletes intervocalic /p/. Mandar allows the segment /p/ to appear freely in intervocalic positions, and its status in that position has been diachronically stable: every known /VpV/ sequence in a lexeme inherited from Proto-Malayo-Polynesian has survived unchanged. The following examples illustrate this fact, drawing reconstructions from Blust and Trussel 2023:

(50) *No reduction of intervocalic /p/*

- Mandar: tapa api sapu  
 PMP: tapa api sapu  
 Gloss: smoke fire wipe

These segmental alternations are also unattested in the functional vocabulary. Mandar has many functional items that are segmentally similar to the alternating X<sup>0</sup>s in (46), like the demon-

strative *itij* ‘that’ (out of sight). No matter the position of this *Dem*<sup>0</sup>, it always takes the disyllabic form [itij] and cannot lose the initial segment /i/.

(51) *No /i/-aphaeresis in the demonstrative itij*

- a. aláŋa<sup>H</sup> ā [DP **ítij**<sup>H</sup> \_\_\_\_].  
 ala-aŋ a? itij \_\_\_\_  
 get-APPL 1ABS that  
 ‘Get that for me.’
- b. aláŋa<sup>H</sup> ā [DP **\*(i)tim** búku<sup>H</sup>].  
 ala-aŋ a? itij buku  
 get-APPL 1ABS that book  
 ‘Get me that book.’

In the same vein, Mandarin has a complementizer /apa?/ ‘because’ that cannot alternate like *apa* ‘what’. Even in the middle of a *φ*, it cannot reduce to *a?* (52).

(52) *No /apa/-reduction in the functional vocabulary*

- póle<sup>H</sup> i **a\*(pa)?** díní<sup>H</sup> i kindó?na<sup>H</sup>.  
 pole i apa? dini i kindo?-na  
 come 3ABS because here 3ABS mother-3GEN  
 ‘She came because her mother was here.’  
 (Sikki et al. 1987:109)

These observations reveal that the segmental alternations in this system are unattested beyond the narrow set of *X*<sup>0</sup>s in table (44). There is no independent evidence to suggest, in any stratum of the language, that segments like /p/ and /i/ can be unlinked, gradiently present, or otherwise liable to delete. In this respect, they are similar to many of the segmental alternations that emerge when functional heads are stranded in English (e.g., there is no general reduction of [wʌd] to [w-/d] and no regular process that deletes /v/ in the functional stratum, given /ʌv/-/ə/ but finite /hæv/-/v/, not \*/ə/). This result raises a first strike against a synchronic phonological analysis.

We can now strengthen the case for suppletion with two further lines of evidence. The first involves a lack of gradience. Some of the auxiliaries that alternate in English are able to take forms that fall phonologically between their short and long extremes when they are not completely deaccented (e.g., /hæv/-/həv/-/əv/-/v/). But these types of intermediary forms are unattested in the Mandarin system in table (44). The table in (53) illustrates: the *Dem*<sup>0</sup>s *de* and *do*, the negator *ndaŋ*, the adverb *de*, and the *wh*-word *a* cannot take forms that are intermediary between their short and long allomorphs.

(53) *No intermediate forms*

| Gloss      | this | that | not    | what |
|------------|------|------|--------|------|
| Attested   | de   | do   | ndaŋ   | a    |
| Unattested | *dje | *djo | *ndjaŋ | *aφa |
| Attested   | di?e | di?o | andiaŋ | apa  |

This fact suggests that the short and long forms of these  $X^0$ s are distinguished categorically in the grammar, rather than forming the endpoints of a synchronic derivational cline that can be partially interrupted by the phonology of focus.

The second argument for the suppletive analysis involves affixation. There are a number of affixes in Mandar that can appear on the alternating  $X^0$ s in table (44). For instance, there is a prefix *ya-* that expones an  $X^0$  in the extended projection of  $N^0$  and can appear above  $\text{Dem}^0$ s in contexts of deixis (cf. Indonesian *yang ini* ‘this one’). When *ya-* appears above the  $\text{Dem}^0$ s *de* and *do*, it forces them to take their short forms (54).

- (54) *Demonstrative + prefix ya-*  
 aláŋa<sup>H</sup>    ǎ    [DP **ja.ɿo**    waláo<sup>H</sup> ʔó<sup>H</sup> ].  
 ala-aŋ    aʔ    ia-do    balao    o  
 get-APPL 1ABS    YA-that mouse there  
 ‘Get me that mouse there’ (pointing to it).

When *de* and *do* host the prefix *ya-*, they must retain their short forms even when stranded. The following example illustrates with *do* in the context of NP-ellipsis:

- (55) *Demonstrative + prefix ya-: No alternation under stranding*  
 aláŋa<sup>H</sup>    ǎ    [DP **já.ɿo**<sup>H</sup>    \_\_\_\_ ʔó<sup>H</sup> ].  
 ala-aŋ    aʔ    ia-do    \_\_\_\_ o  
 get-APPL 1ABS    YA-that    there  
 ‘Get me that (thing) there’ (pointing to it).

Many other affixes force an analogous pattern of suspension with the root  $\sqrt{\text{WHAT}}$ , including *saŋ-* (*saŋ-apa* ‘how many’) and the verbalizers *maŋ-* (*maŋ-apa* ‘do what’), *ti-* (*ti-apa* ‘be what’), and *mi-* (*mi-apa* ‘how’). When  $\sqrt{\text{WHAT}}$  combines with any of these  $X^0$ s, it takes the long allomorph *apa* (56a)—even when it is nonfinal in its  $\phi$  (56b).

- (56) *Affixal freezing: a*  
 a. máápa<sup>H</sup>    o    míeʔ<sup>H</sup>?  
    maŋ-apa    o    mieʔ  
    VERBALIZER-what 2ABS 2PL  
    ‘What are you guys doing?’  
 b. máápa    násaŋ<sup>H</sup> o    míeʔ<sup>H</sup>?  
    maŋ-apa    nasasŋ o    mieʔ  
    VERBALIZER-what every 2ABS 2PL  
    ‘What are all of you guys doing?’

In contrast, on the suppletive analysis there is a natural way to understand this effect. If the short and long forms of the alternating  $X^0$ s are separate vocabulary entries, then they should be able to be independently selected by other  $X^0$ s in the language. For instance, the  $v^0$  that hosts the affix *maŋ-* might select the long allomorph of  $\sqrt{\text{WHAT}}$ , while the  $X^0$  that hosts *ya-* might select the short allomorph of  $\sqrt{\text{THAT}}$ . The result is an analysis that allows us to treat these effects as routine cases of outward-sensitive morphosyntactically conditioned allomorphy (see, e.g., Carstairs 1988,

Embick 2010). On a purely phonological approach, the analysis of this pattern would be much less clear.<sup>8</sup>

(57) *The suppletive analysis: Outward-conditioned allomorphy*

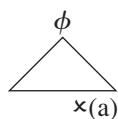
- a.  $\sqrt{\text{WHAT}} \rightarrow \text{apa} / \text{v}^0_{\text{mag-/mi-/ti-}} \text{---}$   
 b.  $\sqrt{\text{THAT}} \rightarrow \text{do} / \text{F}^0_{\text{ya-}} \text{---}$

3.3 *Vocabulary Insertion Occurs in the Phonology*

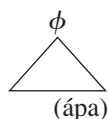
These three considerations—the lack of comparable segmental alternations, the lack of intermediate forms, and the interaction with affixation—suggest that the alternations in table (44) involve suppletion. This conclusion allows us to turn to the theory of VI. I argue that the patterns of top-down suppletion in Mandarin are driven by the Positional Minimality Constraint. The diagram in (58) illustrates the account: the seven  $X^0$ s in table (44) preferentially surface in their monosyllabic allomorphs, the Positional Minimality Constraint bans the appearance of these allomorphs at the right edge of the  $\phi$ , and this constraint thus forces these  $X^0$ s to supplete to their disyllabic allomorphs.

(58) *Top-down suppletion: Output-optimizing*

a. Constraint



b. Repair



This analysis allows us to integrate these seven cases of top-down suppletion in Mandarin with the larger literature on output-optimizing suppletion (Mester 1994, Kager 1996, Tranel 1996, Dolbey 1997, Booij 1998, Perlmutter 1998, Rubach and Booij 2001, Wolf 2008, 2015, Bonet and Harbour 2012, Bennett 2017), or *External Allomorphy* in the terminology of Mascaró 1996, 2007. The literature on this type of suppletion has largely focused on cases of suppletion within the prosodic word, typically involving affixes that supplete around certain types of hosts in response to output phonological constraints. For example, Mandarin bans sequences of identical vowels within the  $\omega$ . This constraint forces such vowel sequences to coalesce at many word-internal morpheme boundaries, as shown with the  $a^0$  *ma-* in the examples in (59b), but it drives a different response in the head  $\text{Appl}^0$ . The unmarked allomorph of  $\text{Appl}^0$  in Mandarin is *-aŋ*, but this head takes the lexically marked allomorph *-ŋaŋ* after verbal roots that end in */a/* (59c).

<sup>8</sup> The most natural way to address this pattern on a synchronic phonological analysis, as I understand it, would be to appeal to the cycle: positing that (a) VI invariably inserts the long forms of the  $X^0$ s in table (44); (b) certain segments in the long forms are prespecified as liable for deletion, or alternatively the long forms themselves are marked as liable to undergo certain processes (Pater 2009, Gouskova 2012); and (c) certain affixal  $X^0$ s force their complements to undergo an opaque cycle of Spell-Out where they are mapped to  $\phi$ s and subjected to the Positional Minimality Constraint, blocking the synchronic reduction of *apa* to *a* at this early stage. The clearest issue for this account is that the Positional Minimality Constraint is transparently enforced over the surface prosody of the clause, forcing the selection of short and long allomorphs at a derivational stage that follows focus fronting and the linearization of second-position elements (section 2).



(59) *Output-optimizing allomorph selection within the word*

a. CONSTRAINT

[<sub>ω</sub> . . . \*aa . . . ]

b. Response 1: Coalescence

**marása maráras mámmis másɿŋ**

**ma-rasa ma-raras ma-ammis ma-asing**

ADJ-tasty ADJ-spicy ADJ-sweet ADJ-salty

‘tasty, spicy, sweet, salty’

c. Response 2: Allomorphy

**sjóan̩ álljan̩ ránnwan̩ aláŋan̩ annáŋan̩ bawáŋan̩**

sio-an̩ alli-an̩ rannu-an̩ ala-an̩ anna-an̩ baba-an̩

order-APPL buy-APPL hope-APPL take-APPL save-APPL bring-APPL

‘order for, buy for, hope for, take for, save for, bring for’

On an output-optimizing analysis, this type of pattern can be understood as follows. When VI applies to a given  $X^0$ , all possible vocabulary entries that could be inserted into that  $X^0$  are compared (Halle and Marantz 1993). These vocabulary entries are ordered into a ranked hierarchy of lexical markedness, and there is a systematic preference to insert the least marked vocabulary entry wherever possible (Kiparsky 1973, Halle and Marantz 1993). This pressure is invariably enforced in the phonology, where it is formalized as the constraint PRIORITY (Mascaró 1996, 2007) (60a). This constraint enforces the allomorph ordering of  $-aŋ > -ŋaŋ$  for the head Appl<sup>0</sup> in Mandar, and when external factors do not interfere, it forces VI of  $-aŋ$  (60b).

(60) *The constraint PRIORITY*

a. PRIORITY

Given an input containing an  $X^0$  that can be realized with the lexically ordered allomorphs  $\{M_1, M_2, \dots, M_n\}$  and a candidate that contains  $M_i$ , assign as many violation marks as the depth of ordering between  $M_i$  and  $M_1$ .

|    |                                                                                                 |          |
|----|-------------------------------------------------------------------------------------------------|----------|
| b. | $\sqrt{\text{BUY-APPL}}$ ‘buy for’                                                              | PRIORITY |
|    |  a. alli-an̩ |          |
|    | b. alli-ŋan̩                                                                                    | *!       |

The central intuition of the output-optimizing analysis is that PRIORITY is a ranked and violable constraint that is evaluated in parallel with other constraints in the phonology. When PRIORITY is ranked above conflicting phonological constraints, it drives patterns of allomorph selection that are not output-optimizing, patterns of phonologically conditioned suppletion that are anti-optimizing (Bonet, Lloret, and Mascaró 2007), and patterns of suppletion that are not phonologically conditioned at all. But when PRIORITY is dominated by conflicting phonological constraints, it becomes possible for output constraints to determine the choice of allomorph that is inserted by VI. This situation, which implicates a ranking of “ $P > M$ ” in the sense of McCarthy and Prince 1993, allows us to derive the allomorphic alternation between  $-aŋ$  and  $-ŋaŋ$  from independently visible output constraints. When the head Appl<sup>0</sup> appears before a root that ends

in /a/, the constraints that restrict sequences of identical vowels (61a) and restrict coalescence (61b) outrank PRIORITY and force the insertion of the lexically marked allomorph -*ɲaɲ* (61c).

(61) *Output optimization drives suppletion*

a. OCP-V

Assign one violation for every sequence of identical vowels within the  $\omega$ .

b. MAX-V

Assign one violation for every input vowel with no output correspondent.

| c. | $\sqrt{\text{TAKE-APPL}}$ ‘take for’ | OCP-V | MAX-V | PRIORITY |
|----|--------------------------------------|-------|-------|----------|
|    | a. ala-aɲ                            | *!    |       |          |
|    | b. ala-ɲ                             |       | *!    |          |
|    | ☞ c. ala-ɲaɲ                         |       |       | *        |

Extending this logic to the system of top-down suppletion, I propose that the alternations in table (44) are driven by the Positional Minimality Constraint in the same way. Focusing on the alternation between *a* and *apa*, I assume that the emergence of the lexically marked allomorph *apa* in  $\phi$ -final positions, shown in example (62b), is forced by a ranking of the Positional Minimality Constraint over PRIORITY.

(62) *Irregular expansion: A-apa*

a. **á**      ɹúa<sup>H</sup>   píllóa<sup>H</sup>?

a      dua   piɲ-loa  
what still   ANTIP-sound  
‘What’s still ringing?’

b. **ápa**<sup>H</sup>   píllóa<sup>H</sup>?

a      piɲ-loa  
what   ANTIP-sound  
‘What’s ringing?’

Tableaux (63) and (64) illustrate this account. When  $\sqrt{\text{WHAT}}$  is nonfinal in the  $\phi$ , it is not subjected to the Positional Minimality Constraint. In this context, PRIORITY can force the insertion of the unmarked allomorph *a*. This configuration emerges when  $\sqrt{\text{WHAT}}$  is followed by a  $\phi$ -internal second-position element like *dua* ‘still’ (62a), which is parsed into a minimal  $\phi$  with  $\sqrt{\text{WHAT}}$  by the ranking of SUBCAT over MATCH( $\phi$ ,XP).

(63) *PRIORITY in the system of top-down suppletion*

| $\sqrt{\text{WHAT}}$ $\sqrt{\text{STILL}}$ ANTIP- $\sqrt{\text{SOUND}}$ ‘What’s still ringing?’                     | SUB | POS-MIN | PRI | M( $\phi$ ,XP) |
|---------------------------------------------------------------------------------------------------------------------|-----|---------|-----|----------------|
| ☞ a. $\{\phi[\omega(\acute{a})][\omega(\acute{d}úa)]\} \{\phi[\omega(\acute{p}íl)][\omega(\acute{l}óa)]\}$          |     |         |     | *              |
| b. $\{\phi[\omega(\acute{a}pa)][\omega(\acute{d}úa)]\} \{\phi[\omega(\acute{p}íl)][\omega(\acute{l}óa)]\}$          |     |         | *!  | *              |
| c. $\{\phi[\omega(\acute{a}pa)]\} \{\phi[\omega(\acute{d}úa)]\} \{\phi[\omega(\acute{p}íl)][\omega(\acute{l}óa)]\}$ | *!  |         | *   |                |

In the contexts where  $\sqrt{\text{WHAT}}$  appears at the right edge of the  $\phi$ , it must respond to the Positional Minimality Constraint. As this constraint outranks PRIORITY, the Positional Minimality Constraint can force VI to insert the marked allomorph *apa*.

(64) *Positional minimality* > *PRIORITY*

| $\sqrt{\text{WHAT}}$ ANTIP- $\sqrt{\text{SOUND}}$ ‘What’s ringing?’                    | POS-MIN | PRIORITY |
|----------------------------------------------------------------------------------------|---------|----------|
| a. $\{\phi[\omega(\acute{a}pa)]\} \{\phi[\omega(p\acute{í}l)] [\omega(l\acute{o}a)]\}$ |         | *        |
| b. $\{\phi[\omega(\acute{a})]\} \{\phi[\omega(p\acute{í}l)] [\omega(l\acute{o}a)]\}$   | *!      |          |

The result is an analysis that captures the distribution of the short and long allomorphs in this system in a precise and restrictive way. First, it integrates seven patterns of suppletion under a single rubric: certain  $X^0$ s take monosyllabic forms in  $\phi$ -medial positions and disyllabic forms at the right edge of the  $\phi$ . Second, it explains the identity of the positions that trigger alternations (the positions that enforce the driving constraint) and the prosodic shapes of the alternations that occur ( $((\sigma) \rightarrow (\acute{\sigma}\sigma))$ ). Third, it makes correct predictions about the types of  $X^0$ s that can and cannot supplete (see also Kayne 2019): top-down suppletion can only ever target  $X^0$ s whose unmarked forms are monosyllabic. Fourth, it allows us to understand how these patterns of suppletion might march in lockstep with phonological alternations that have the same distribution and prosodic effect: epenthesis, hiatus nonresolution, and suppletion are triggered by a single constraint in the phrasal phonology. The scale of this success suggests that the output-optimizing analysis is fundamentally on the right track for these alternations in Mandarin, and as a result, we arrive at a new empirical argument that output-oriented phonological constraints can guide the operation of VI.

### 3.4 The Case for Output Optimization

This optimizing analysis contrasts with an alternative approach, first developed by Paster (2006) and Bye (2007), which holds that all cases of phonologically conditioned suppletive allomorphy should be handled in terms of selection. The initial case for this alternative is built on the logic of parsimony: selectional analyses are necessary for many patterns of allomorphy, so the theory should generalize this type of approach to all cases of allomorphy and eschew output optimization entirely. On this approach and many later extensions (Embick 2010, Gouskova and Bobaljik to appear, Kalin 2020, Stanton 2022), the alternations in this system would be captured by positing two selectional frames for each  $X^0$ : one associated with the long allomorph that selects the right edge of the  $\phi$  and one associated with the short allomorph that appears elsewhere.

(65) *Rejected: The selectional theory*

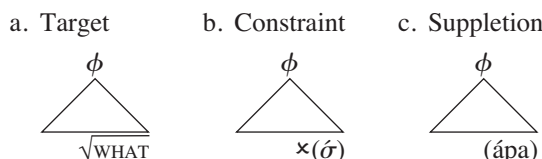
- a.  $\sqrt{\text{WHAT}} \rightarrow \text{apa} / \text{ — } )_{\phi}$
- b.  $\sqrt{\text{WHAT}} \rightarrow \text{a}$

There are three reasons to reject an account of this type. The first argument is conceptual. There are many patterns of allomorph selection that cannot be derived by output constraints, including some that seem to run against plausible output-oriented constraints in the phonology

(Booij 1998). But there are many other systems that operate in the phonology and resist analysis in output-optimizing terms, such as the circular chain shifts that emerge in systems of tone sandhi (Moreton 1999, Barrie 2006). Output optimization cannot be reasoned out of existence on the grounds that selection is needed for cases of this sort—no more than the morphological module can be reasoned out of existence on the grounds that the theory independently requires the operation Merge (cf. Collins and Kayne 2023). The verdict on output optimization should be decided instead on two grounds: the top-down requirement for logical consistency in the theory of VI and the bottom-up mandate for explanatory adequacy in the domain of suppletion.

The second argument against this type of account emerges from the logic of late insertion. We have seen that there are cases where the output of VI becomes predictable from constraints that operate in the phonology: when the root  $\sqrt{\text{WHAT}}$  appears at the right edge of the  $\phi$  (66a), the Positional Minimality Constraint will ban its short allomorph (66b) and force out its long form (66c) in the same way that it forces other monosyllabic  $X^0$ s to undergo segmental repairs in the same position.

(66) *Phonological constraints*  $\rightarrow$  *Insertion after morphology*



On a selectional analysis, it would be necessary to capture these patterns by positing seven disjunctive selectional frames that coincidentally shared the same shape and provided suppletion with the same distribution and prosodic shape as processes in the phonology: in other words, it would be necessary to duplicate the effects of an independently visible output constraint (Kisberberth 1970). The desire to avoid this type of duplication played a central role in the debates that led to the elimination of morpheme structure constraints (Stanley 1967, Hooper 1972, Clayton 1976) and gave rise to the theory of underspecification (Kiparsky 1982, Archangeli 1984, Steriade 1987). The same drive, in turn, underpins the theory of late insertion in DM. Halle and Marantz (1993) observe that there are cases in which patterns of exponence are conditioned by constraints that are plausibly enforced only in postsyntactic modules like the morphology. To provide one example, the genitive agreement paradigm in classical Mandar shows a syncretism between 3SG and 3PL (67a). This syncretism can be derived by positing a constraint on the cooccurrence of the features 3GEN and PL in the morphological module in this language (67b), forcing an operation of impoverishment to apply and remove number features in this context (67c). As this process of impoverishment would then determine the form that appears in this context, such an account would need to posit an operation of late insertion that applied in the morphology or a later derivational stage.

(67) *Morphological constraints*  $\rightarrow$  *Insertion after syntax*

|         |         |         |         |         |         |         |   |
|---------|---------|---------|---------|---------|---------|---------|---|
| a. loka | [-ʔu    | / -ʔi   | / -mu   | / -meʔ  | / -na   | / -na   | ] |
| banana  | 1SG.GEN | 1PL.GEN | 2SG.GEN | 2PL.GEN | 3SG.GEN | 3PL.GEN |   |

‘my / our / your (SG / PL) / her / their bananas’

(Pelenkahu, Muthalib, and Sangi 1983:110)

b. Morphological constraint

\*[3GEN, PL]

c. Morphological rule

PL  $\rightarrow$   $\emptyset$  / [3GEN, \_\_\_\_]

The same logic leads directly to the case that VI operates in the phonology: when patterns of allomorph selection become predictable at a certain derivational stage, it is parsimonious and explanatory to handle allomorph selection there. To reject the case that VI operates in the phonology is thus to reject the logic that justifies late insertion. There is no internally consistent theory that posits a morphological operation of VI to capture generalizations that emerge in the morphology and then shirks responsibility for capturing all parallel generalizations that emerge in the phonology. Rather, the foundational logic of late insertion leads directly to the output optimization account.

The third argument against a selectional account lies with the mandate for explanatory adequacy. If the alternations in the system of top-down suppletion were analyzed in terms of selection, we would have no way to explain the various patterns that we have derived above. To abandon this success and adopt a selectional analysis would be to commit to reaching no farther than a baseline level of description for a phenomenon that is clearly embedded within the synchronic grammar of Mandarin. A purely selectional theory of these patterns of allomorph selection would thus fail to rise to the analytical standard that justifies and guides the generative enterprise (Chomsky 1965). In pursuit of genuine explanation, rather, output optimization is required.

#### 4 Allomorphic Vacillation

These architectural conclusions suggest that the postsyntax must have a shape that differs slightly from that which is typically hypothesized in DM (Embick and Noyer 2001, Arregi and Nevins 2012). Keeping the structure of the morphological module intact, I propose the revised model in (68): one in which VI is shifted past the tail end of the morphological module to be handled in the phonology. This model allows all of the standing syntactic generalizations on VI to persist in that module, plausibly enforced by ranked and violable constraints that read off of the morpho-syntactic structure of the input, and it opens up a natural path for VI to be guided by output-oriented constraints.

(68) *The architecture of the grammar*

| Syntax                             | Morphology                             | Phonology               |
|------------------------------------|----------------------------------------|-------------------------|
| Syntactic X <sup>0</sup> -movement | Morphological X <sup>0</sup> -movement | Mapping to prosody      |
| Agree-Link                         | Agree-Copy                             | Linearization           |
| Ellipsis Licensing                 | Fission                                | Ellipsis implementation |
| XP-movement                        | Impoverishment                         | Vocabulary Insertion    |

This reformulation opens up a final question regarding the amount of phonological context that is visible to VI. As the Positional Minimality Constraint operates over the surface prosody, I have argued that it applies in the global and parallel phonological evaluation in which all prosodic structure is built. On this analysis, VI should target the  $X^0$ s in this system in a calculus that has access to the full phonological content of the clause. Our final task is thus to weigh the evidence that VI can operate in a global phonological calculus of this type—and strengthen the case for output optimization along the way.

#### 4.1 Syllable Lifting

The output-optimizing analysis holds that the patterns of top-down suppletion in Mandarin are driven by output constraints, rather than the lexical requirements of the alternating  $X^0$ s themselves. As a result, this type of analysis should predict—as Embick (2010) points out—that these patterns of suppletion might be suspended in the positions where they usually occur if the larger phonological context allows the driving constraint to be resolved in a more optimal way.

To test this hypothesis, we can begin by establishing that the Positional Minimality Constraint does indeed have an additional repair of this type. Mandarin has a set of agreement clitics that surface in second position and typically follow the initial  $\phi$ . Their parse is shown in example (69a): the 3<sub>ABS</sub> clitic *i* follows the  $\phi$ -internal *nasaj* and falls after the  $\phi$ -final H tone. But this parse changes when the 3<sub>ABS</sub> clitic follows a  $\phi$ -internal element that is monosyllabic, like *to* ‘also’. In that context, the  $\phi$ -final monosyllable does not respond to the Positional Minimality Constraint in its usual way (here, V?-epenthesis). Instead, the  $\phi$ -final monosyllable retains its usual shape and the  $\phi$ -final H tone shifts one syllable to the right, falling after the 3<sub>ABS</sub> clitic (69b).

##### (69) *Absolute clitics: A change in parse*

- a. lóppa? násaŋ<sup>H</sup> i.  
     loppa? nasaj i  
     hot every 3<sub>ABS</sub>  
     ‘Every one of them is hot.’
- b. lóppa? tó i<sup>H</sup>.  
     loppa? to i  
     hot also 3<sub>ABS</sub>  
     ‘It’s also hot.’

The same effect obtains when  $\phi$ -final monosyllabic elements appear before apparently unfooted syllables of any type. The adverb *to*, for instance, immediately precedes the predicate in clauses where absolute DPs undergo *wh*-movement (absolute clitics disappear in an *anti-agreement effect*; Ouhalla 1993, Baier 2018, Brodtkin 2021). When the predicate begins in a stressed syllable, *to* shows V?-epenthesis (70a). But when the predicate begins in an unstressed syllable, this repair is suspended (70b). At the same time, the  $\phi$ -final H tone shifts one syllable to the right.

(70) *The suspension of Positional Minimality*

- a. ikátʃoʔ **tóʔo<sup>H</sup>** yúru<sup>H</sup>.  
     ikatʃoʔ to      guru  
     NAME also teacher  
     ‘Kacho’ is also a teacher.’
- b. ikátʃoʔ **tó** yu<sup>H</sup>rúnna<sup>H</sup>.  
     ikatʃoʔ to      guru-na  
     NAME also teacher-3GEN  
     ‘Kacho’ is also one of her teachers.’

The same repair extends across the system of Positional Minimality. When the P<sup>0</sup> *suŋ* ‘out’ is stranded before a disyllabic argument by DP-ellipsis, it forms a  $\phi$  and shows V?-epenthesis (71a). But when stranded before a trisyllabic argument, it shows no epenthesis. Instead, the first syllable of that argument falls into the preceding  $\phi$  (71b).

(71) *The broader system of suspension*

- a. bémme<sup>H</sup> i      **súʔun<sup>H</sup>** dóiʔ<sup>H</sup>.  
     bemme i      suŋ      doiʔ  
     fall      3ABS out      money  
     ‘The money fell out.’
- b. bémme<sup>H</sup> i      **sún** do<sup>H</sup>íʔmu<sup>H</sup>.  
     bemme i      suŋ doiʔ-mu  
     fall      3ABS out money-2GEN  
     ‘Your money fell out.’

The same pattern emerges with X<sup>0</sup>s of the shape CVV(C). When the P<sup>0</sup> *nauŋ* ‘down’ is stranded before a disyllabic subject, it retains its underlying sequence of vowels (72a). But when it appears before a trisyllabic subject, it coalesces to the form *noŋ*. This reduction would normally violate the Positional Minimality Constraint, but this form is rescued by pulling the initial syllable of the following subject into its  $\phi$  (72b). The result is an interaction that requires the simultaneous application of coalescence and movement: a case of irreducible parallelism, in the terminology of Adler and Zymet (2021).

(72) *Coalescence and a second repair*

- a. támbus<sup>H</sup> i      **náuŋ<sup>H</sup>** állo<sup>H</sup>.  
     tambus i      nauŋ      allo  
     set      3ABS down sun  
     ‘The sun went down.’
- b. támbus<sup>H</sup> i      **nóŋ** al<sup>H</sup>lóna<sup>H</sup>.  
     tambus i      nauŋ      allo-na  
     set      3ABS down sun-3GEN  
     ‘The sun went down.’

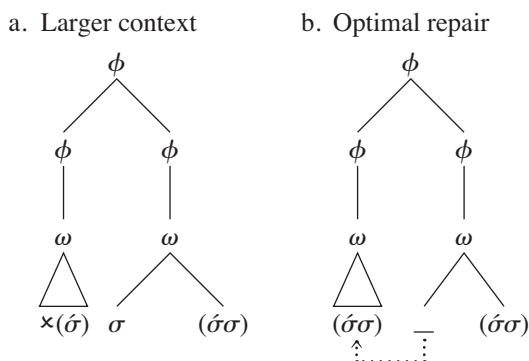
CVV(C) nouns trigger the same effect. The noun *sia* ‘salt’ shows hiatus when it appears as an internal argument before a disyllabic applied argument (73a). But it shows gliding and pulls in a syllable before a trisyllabic applied argument (73b).

(73) *A systematic preference*

- a. wánne<sup>H</sup> i sía<sup>H</sup> pétʃaʔ<sup>H</sup>.  
 u-anna-i i sia petʃaʔ  
 1ERG-add-APPL 3ABS salt porridge  
 ‘I added salt to the porridge.’
- b. wánne<sup>H</sup> i sjá wa<sup>H</sup>róʔboʔ<sup>H</sup>.  
 u-anna-i i sia baroʔboʔ  
 1ERG-add-APPL 3ABS salt corn.porridge  
 ‘I added salt to the corn porridge.’

These patterns reveal that there is an additional repair to the Positional Minimality Constraint that is preferred to those we have seen so far. This is one that creates disyllabic feet at the right edge of the  $\phi$  by drawing in otherwise-unfooted syllables from the left edge of the following  $\phi$ , in a case of phonological movement driven by the needs of the target position (akin to the types of syntactic displacement that are driven by Attract (Chomsky 2001)). This movement is preferred over all other strategies that might be leveraged to satisfy that need (in a kind of phonological Move-over-Merge). I will refer to this final repair as *Syllable Lifting*. Its shape is sketched in diagram (74).

(74) *Syllable Lifting*



The preference for Syllable Lifting suggests that the phonology of Mandarin prefers to avoid the segmental repairs to the Positional Minimality Constraint by breaking the usual isomorphism between syntactic and prosodic constituents. Drawing a syllable across the boundaries of  $\omega$ s and  $\phi$ s creates syntax-prosody misalignment in four positions: at the right edge of the first  $\omega$ , at the right edge of the first  $\phi$ , at the left edge of the second  $\omega$ , and at the left edge of the second  $\phi$ . We can understand this effect to emerge from the ranking of the constraints that favor monosyllabism (ONSET and DEP) over the constraints that demand isomorphism between syntactic and



prosodic constituents ( $\text{MATCH}(\phi, \text{XP})$  and  $\text{MATCH}(\omega, X^0)$  (75a)). The result is shown in tableau (75b): focusing on example (72b),  $X^0$ s like the  $P^0$  *naung* are forced to show coalescence at the right edge of the  $\phi$  and satisfy the Positional Minimality Constraint by drawing in a syllable from the following  $\phi$ .

(75) *The derivation of Syllable Lifting*

a.  $\text{MATCH}(\omega, X^0)$

Assign one violation for every  $\omega$  in an output phonological representation P whose edges are not aligned with the edges of a corresponding  $X^0$  in an input syntactic representation S.

b. *Syllable Lifting:  $\text{ONSET} > \text{MATCH}$*

| [ <sub>TP</sub> $\sqrt{\text{SET}}$ 3ABS $\sqrt{\text{DOWN}}$ $\sqrt{\text{SUN}}$ -3GEN] ‘The sun set.’                                                     | ONS | $M(\omega, X^0)$ | $M(\phi, \text{XP})$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------|----------------------|
| ☞ a. $\{\phi\{\phi_{[\omega]}(\text{tám} \text{bus})\}\} \text{ i } \{\phi_{[\omega]}(\text{nó} \text{ŋal})\}\} \{\phi_{[\omega]}(\text{lóna})\}\}$         |     | **               | **                   |
| b. $\{\phi\{\phi_{[\omega]}(\text{tám} \text{bus})\}\} \text{ i } \{\phi_{[\omega]}(\text{ná} \text{uŋ})\}\} \{\phi_{[\omega]}(\text{al}(\text{lóna}))\}\}$ | *!  |                  |                      |

4.2 *The Suspension Effect*

With this repair now established, we can return to the system of top-down suppletion. We have already seen that the root  $\sqrt{\text{WHAT}}$  suppletes to the long allomorph *apa* at the right edge of a  $\phi$  when the following  $\phi$  begins with a stressed syllable, like the prefix *píŋ-* in example (40b) or the prefix *máŋ-* in example (76a). But this root takes its short allomorph at the right edge of a  $\phi$  and triggers Syllable Lifting when the following  $\phi$  begins with an unstressed syllable, like the future prefix *na-* in example (76b). In the contexts that allow Syllable Lifting, in other words, the pattern of top-down suppletion is suspended for this  $X^0$ . I will refer to this pattern as the *suspension effect*.

(76) *The suspension effect: Context 1*

a. **ápa**<sup>H</sup> máppaké?de?<sup>H</sup> i?

a      maŋ-pake?de? i  
what AF-awaken      3ACC  
‘What woke him up?’

b. **á**      na<sup>H</sup>máppaké?de?<sup>H</sup> i?

a      na-maŋ-pake?de? i  
what FUT-AF-awaken      3ACC  
‘What will wake him up?’

The same pattern holds across the system of top-down suppletion: in the phonological contexts that allow Syllable Lifting, the heads in table (44) exceptionally take their short allomorphs at the right edge of the  $\phi$ . The examples in (77) show the same fact with the root  $\sqrt{\text{WHICH}}$ , using monomorphemic predicates of different phonological shapes. When  $\sqrt{\text{WHICH}}$  appears before the disyllabic *keccu* ‘small’, whose initial syllable is stressed, it suppletes to the long allomorph *inna*

(77a). But when it appears before the trisyllabic *karambo* ‘far’, whose initial syllable is unstressed, it triggers Syllable Lifting and surfaces as the unmarked allomorph *na* (77b).

(77) *The suspension effect: Context 2*

- a. **ínna**<sup>H</sup> kɛtʃuʔ<sup>H</sup>?  
 na kɛtʃuʔ  
 which small  
 ‘Which one is smaller?’
- b. **ná** ka<sup>H</sup>rámbo<sup>H</sup>?  
 na karambo  
 which far  
 ‘Which one is farther away?’

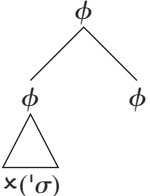
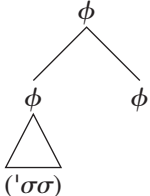
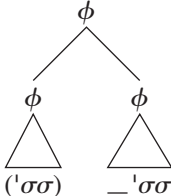
The examples in (78) show the same pattern with one of the Dem<sup>0</sup>s in this system. When a Dem<sup>0</sup> is stranded in the position of an internal argument before an applied argument that begins in a stressed syllable, it suppletes to its long allomorph (78a). But when a Dem<sup>0</sup> is stranded in this position before an applied argument that begins in an unstressed syllable, it triggers Syllable Lifting and takes its unmarked allomorph (78b).

(78) *The suspension effect: More contexts*

- a. nalattʃári<sup>H</sup> i **índi**<sup>H</sup> \_\_\_ yúru<sup>H</sup> é<sup>H</sup>.  
 na-latʃari i ndi \_\_\_ guru e  
 3ERG-throw at 3ABS this teacher here  
 ‘She threw this (thing) here at the teacher.’
- b. nalattʃári<sup>H</sup> i **ndí** \_\_\_ yu<sup>H</sup>rúnna<sup>H</sup> é<sup>H</sup>.  
 na-latʃari i ndi \_\_\_ guru-na e  
 3ERG-throw at 3ABS this teacher-3GEN here  
 ‘She threw this (thing) here at her teacher.’

The suspension effect thus reveals another layer of phonological conditioning in the system of top-down suppletion. The Positional Minimality Constraint bans degenerate feet at the right edge of the  $\phi$  (79a), and in this position certain X<sup>0</sup>s typically supplete (79b). But this suppletion is a last resort: when the larger context allows for an alternative repair, these X<sup>0</sup>s respond to the Positional Minimality Constraint by drawing in an erstwhile-unfooted syllable from the following  $\phi$  (79c).

(79) *Resolving the Positional Minimality Constraint*

- a. Constraint
- 
- b. Last resort
- 
- c. Optimal repair
- 

The result is that the system of top-down suppletion shows the property that Embick (2010) terms *allomorphic vacillation*: when the global phonological context allows an output constraint to be resolved locally via suppletion or globally by an alternative repair, this system mandates the global repair (Syllable Lifting) and suppletion suspends in the context where it would usually occur (the edge of the  $\phi$ ). Embick (2010) observes that this type of interaction is difficult to find in systems of output-optimizing suppletion that operate at the level of the  $\omega$ , where the forces of lexicalization drive generalization across paradigms (Mester 1994, Albright 2002) and the emergence of pseudocyclic effects (Dressler 1985, Steriade 1999, 2008). But in the system of top-down suppletion, we can see that a global type of allomorphic vacillation does exist.

The initial force of this conclusion is to strengthen the case for output optimization. The suspension effect reveals that there is a second context in which suppletion marches in lockstep with the segmental repairs to the Positional Minimality Constraint. On a selectional analysis, this symmetry would be accidental, unexpected, and impossible to state. But on the output-optimizing analysis, we can capture this symmetry in a manner that is precise, elegant, and well-integrated with the larger phonological system of the language. The tableau in (80) illustrates the basic account: the constraint PRIORITY, which favors VI of the monosyllabic allomorphs in this system, must also be ranked alongside DEP and ONSET above MATCH( $\phi$ ,XP) and MATCH( $\omega$ ,X<sup>0</sup>). The result is a systematic preference to suspend the usual patterns of suppletion in the contexts that allow Syllable Lifting and resolve the Positional Minimality Constraint through movement of an erst-while-unfooted syllable from the following  $\phi$ .

(80) *Syllable Lifting*: ONSET > ALIGN

| [ <sub>CP</sub> $\sqrt{\text{WHICH}}$ . . . [ <sub>PREDP</sub> $\sqrt{\text{FAR}}$ . . . ]]                 | ‘Which is farther?’ | PRI | M( $\omega$ ,X <sup>0</sup> ) | M( $\phi$ ,XP) |
|-------------------------------------------------------------------------------------------------------------|---------------------|-----|-------------------------------|----------------|
| ☞ a. { $\phi$ [ <sub><math>\omega</math></sub> (náka)]} { $\phi$ [ <sub><math>\omega</math></sub> (rámbo)]} |                     |     | **                            | **             |
| b. { $\phi$ [ <sub><math>\omega</math></sub> (inna)]} { $\phi$ [ <sub><math>\omega</math></sub> ka(rámbo)]} |                     | *!  |                               |                |

### 4.3 The Available Context

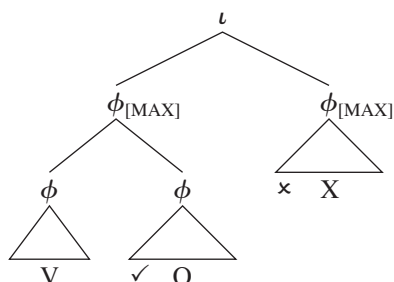
The existence of Syllable Lifting also allows us to study the scale of the phonological context that is visible to VI. Syllable Lifting is constrained by a specific prosodic boundary that defines the domains for many other processes in the phrasal phonology of Mandar (Brodкин to appear). For instance, Mandar has a process of lenition that reduces intervocalic /b d ʈ g/ to [w ɿ j ɿ], and like Syllable Lifting, this process can apply between constituents that form separate minimal  $\phi$ s. Example (81) illustrates its distribution: in a three-word clause of the order VOX (X = adjunct), this type of lenition must apply between the V and the O and cannot apply between the O and the X.

(81) *Lenition*  $\rightarrow \{V O\} \{X\}$

|                                       |                      |         |                      |
|---------------------------------------|----------------------|---------|----------------------|
| [ V                                   |                      | O ]     | [ X ]                |
| mám báwa <sup>H</sup> i               | ɿoʔ áju <sup>H</sup> | di      | yénaʔ <sup>H</sup> . |
| maŋ-baba                              | i                    | doʔaʈʂu | di genaʔ             |
| ANTIP-bring 3ABS vegetable in earlier |                      |         |                      |
| ‘She brought vegetables earlier.’     |                      |         |                      |

In Brodtkin to appear, I take this restriction and several others to suggest that strings of the order VOX show the prosodic constituency in (82): the V and O are parsed into a large prosodic constituent that provides the domain for intervocalic lenition, and this large constituent excludes the X. I argue further that this constituent is an instance of the category  $\phi$  that is directly dominated by the intonational phrase ( $\iota$ ; Ito and Mester 2007, Selkirk 2009)—in other words, a *maximal*  $\phi$  (Ito and Mester 2013).

(82) *Maximal  $\phi$ s and lenition*



The process of Syllable Lifting has the same distribution as lenition. In three-word clauses of the order VSO, lenition must apply between the S and the O (83a). In the same context, Syllable Lifting must also occur between the S and the O when the S can take a monosyllabic form. When a Dem<sup>0</sup> is stranded in the position of the S in a clause of this type, then, it retains its short allomorph and triggers Syllable Lifting from the O (83b).

(83) *Maximal  $\phi$ s  $\rightarrow$   $\checkmark$  Syllable Lifting*

- a.  $\{\phi_{[MAX]}\}$  V S O  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$   
néte<sup>H</sup> i wálo<sup>H</sup> jarámmi<sup>H</sup> ó<sup>H</sup>.  
na-itai i balao ċarammi<sup>ŋ</sup> o  
3ERG-look.for 3ABS mouse mirror there  
‘The mouse there is looking for the mirror.’
- b.  $\{\phi_{[MAX]}\}$  V S O  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$   
néte<sup>H</sup> i jó \_\_\_\_ ja<sup>H</sup>rámmi<sup>H</sup> ó<sup>H</sup>.  
na-itai i do \_\_\_\_ ċarammi<sup>ŋ</sup> o  
3ERG-look.for 3ABS that mirror there  
‘That (thing) there is looking for the mirror.’

At the junctures that ban lenition, in turn, Syllable Lifting is also blocked. In VOX clauses, for instance, Syllable Lifting cannot apply between the O and the X.

(84) *Across maximal  $\phi$ s  $\rightarrow$   $\times$  Syllable Lifting*

- $\{\phi_{[MAX]}\}$  V O  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$   
nálli<sup>H</sup> i .ií?o<sup>H</sup> \_\_\_\_ dirwambóni<sup>H</sup> ó<sup>H</sup>.  
na-alli i do \_\_\_\_ diruambóni o  
3ERG-buy 3ABS that a.while.ago there  
‘She bought that (thing) there a while ago.’

In Brodtkin to appear, I observe that the distribution of lenition is changed in clauses that contain additional prosodic words. In Mandar, the maximal  $\phi$  obeys a constraint on maximal weight: it cannot contain more than three words. As a result, the S and the O are parsed into a single maximal  $\phi$  in three-word clauses of the order VSO, allowing lenition between them (85a), but they are parsed into separate maximal  $\phi$ s, blocking lenition, when an additional  $\omega$  is inserted in the O (85b).

(85) *Maximal  $\phi$ s  $\rightarrow$  Constraints on maximal weight*

- a.  $\{\phi_{[MAX]}\}$  V S O  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$   
 natínro<sup>H</sup> i .ɿo ʔája<sup>H</sup> ʔúru<sup>H</sup> ó<sup>H</sup>.  
 na-tinro i do gaʒa guru o  
 3ERG-chase 3ABS that elephant teacher there  
 ‘That elephant there is chasing the teacher.’
- b.  $\{\phi_{[MAX]}\}$  V S  $\{\phi_{[MAX]}\}$  O  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$   
 natínro<sup>H</sup> i .ɿo ʔája<sup>H</sup> gurúna<sup>H</sup> ikátʃo? ó<sup>H</sup>.  
 na-tinro i do gaʒa guru-na ikatʃo? o  
 3ERG-chase 3ABS that elephant teacher-3GEN NAME there  
 ‘That elephant there is chasing Kacho’s teacher.’

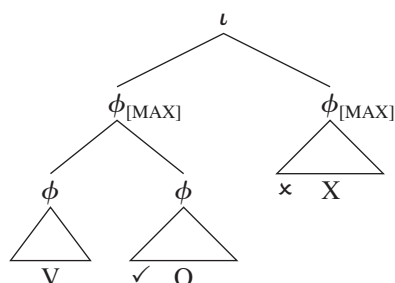
This manipulation affects the distribution of Syllable Lifting in the same way. The examples in (86) illustrate with VSO clauses in which a Dem<sup>0</sup> is stranded in the position of the S. When the VSO string contains three  $\omega$ s in this context, the Dem<sup>0</sup> in the position of the S must take its monosyllabic allomorph and trigger Syllable Lifting (86a). But when another word is added in the O, the Dem<sup>0</sup> in the position of the S can no longer trigger Syllable Lifting and must then supplete to its longer form (86b).

(86) *Weight constraints  $\rightarrow$  Influence Syllable Lifting*

- a.  $\{\phi_{[MAX]}\}$  V S O  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$   
 natínro<sup>H</sup> i .ɿo \_\_\_\_ ʔu<sup>H</sup>rúna<sup>H</sup> ó<sup>H</sup>.  
 na-tinro i do \_\_\_\_ guru-na o  
 3ERG-chase 3ABS that teacher-3GEN there  
 ‘That (thing) there is chasing his teacher.’
- b.  $\{\phi_{[MAX]}\}$  V S  $\{\phi_{[MAX]}\}$  O  $\{\phi_{[MAX]}\}$  X  $\{\phi_{[MAX]}\}$   
 natínro<sup>H</sup> i .ɿʔo<sup>H</sup> \_\_\_\_ gurúna<sup>H</sup> ikátʃo? ó<sup>H</sup>.  
 na-tinro i do \_\_\_\_ guru-na ikatʃo? o  
 3ERG-chase 3ABS that teacher-3GEN NAME there  
 ‘That (thing) there is chasing Kacho’s teacher.’

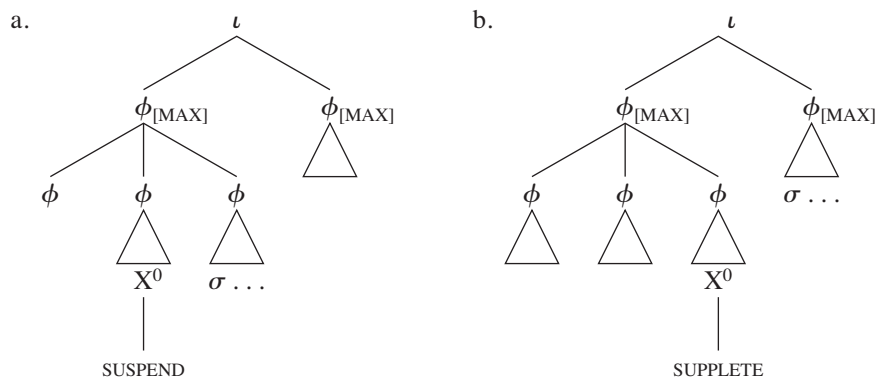
We can thus conclude that Syllable Lifting is conditioned by the same high-level prosodic constituency as lenition: it operates between minimal  $\phi$ s that are parsed into the same maximal  $\phi$  and cannot apply between  $\phi$ s that are parsed into distinct maximal  $\phi$ s. The following diagram illustrates this fact, building from the VOX clause in (84):

(87) *Syllable Lifting*  $\rightarrow$  *Sensitive to the maximal  $\phi$*



We thus arrive at a final layer of phonological conditioning in the system of top-down suppletion. When VI targets one of the  $X^0$ s in this system, it cannot simply inspect the position of the  $X^0$  in its  $\phi$  and the metrical status of the leftmost syllable in the  $\phi$  that follows. Rather, it must also check the way in which the triggering  $\phi$  and the following  $\phi$  are organized into higher prosodic domains. When the two form a single maximal  $\phi$ , suppletion is called off in favor of Syllable Lifting. When the two are parsed into separate maximal  $\phi$ s, Syllable Lifting is banned and suppletion is forced (88).

(88) *The calculus of top-down suppletion*



This conclusion suggests that VI must target the  $X^0$ s in this system at a derivational stage that contains information about the full prosodic organization of the clause. To detect the distribution of maximal  $\phi$ s, more specifically, it must see the distribution of minimal  $\iota$ s—which are typically held to be built in correspondence with CPs in the framework of Match Theory (Selkirk 2009, 2011). On the hypothesis that VI operates within the parallel and global phonological evaluation where prosodic structure is built, this is a natural result.

This footwork allows us to return to the theory of VI. In DP-DP ditransitives in Mandarin, the internal argument (O) precedes the applied argument (D). Despite this fact, it is clear that the O does not c-command the D in clauses of this shape. This fact can be seen in the system of variable binding, building from our earlier discussion of *nasay* ‘all, every’ in section 2. Following Reinhart (1983), in Brodtkin 2022 I argue that variable binding between clausemate arguments in Mandarin is constrained by c-command. As a result of this restriction, VSOD clauses allow the

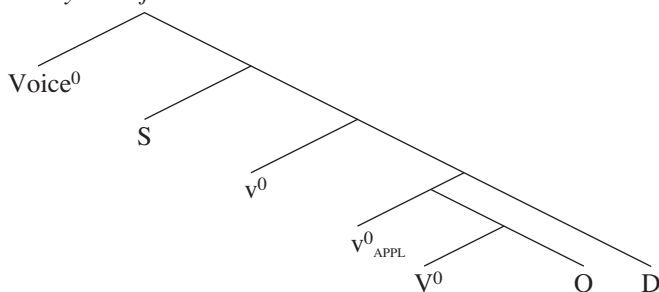
patterns of variable binding in (89): when the S is quantified, it can bind into the final D (89a), but when the O is quantified, it cannot (89b).

(89) *Internal arguments originate beneath applied arguments*

- a. Na-kiring-ang nasang<sub>i</sub> i [<sub>S</sub> panulis<sub>i</sub> \_\_\_\_ ] buku [<sub>D</sub> editor-na<sub>i,j</sub> ].  
 3ERG-send-APPL every 3ABS author book editor-3GEN  
 ‘Every author<sub>i</sub> sent her<sub>i,j</sub> editor a book.’
- b. U-kiring-ang nasang<sub>i</sub> i [<sub>O</sub> buku<sub>i</sub> \_\_\_\_ ] [<sub>D</sub> panulis-na<sub>\*i,j</sub> ].  
 1ERG-send-APPL every 3ABS book author-3GEN  
 ‘I sent its<sub>\*i,j</sub> author every book<sub>i</sub>.’

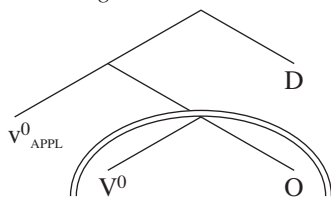
This asymmetry reveals the following hierarchical asymmetry: in the VSOD string, the S c-commands the D and the O does not. The result is the following syntax: the D sits in a rightward specifier of Appl<sup>0</sup> and the O sits beneath it (Brodкин to appear).

(90) *The syntax of the Mandar VoiceP*



Against this backdrop, we can now test the relevance of c-command to the system of top-down suppletion. Bobaljik (2000) proposes that VI applies in a cyclic manner and targets X<sup>0</sup>s one at a time as it moves up through its domain of application. The result is a restriction on phonological accessibility: when VI targets a given X<sup>0</sup>, it should only access the phonological content of the constituents that fall in the c-command domain of that X<sup>0</sup>. The simplest implementation of this hypothesis thus delivers the following prediction: when VI targets the O in a Mandar clause of the order VOD, VI should not be able to see the phonological content of the following D. This predicted restriction on phonological accessibility is sketched with the double line in tree (91).

(91) *Phonological evaluation in the O: Constrained by c-command?*



Example (92) sets up the test case: a Dem<sup>0</sup> is stranded in the position of the O before the trisyllabic D *guru'u* ‘my teacher’. If the Dem<sup>0</sup> were targeted by VI at a derivational stage that

had no access to the phonological shape of the D, then it should be unable to trigger Syllable Lifting and should supplete. This prediction is incorrect: in this context, the  $\text{Dem}^0$  must take its short allomorph and trigger Syllable Lifting.

(92) *Phonological context  $\neq$  C-command domain*

|                           |                                            |      |      |   |                                               |                           |                  |  |
|---------------------------|--------------------------------------------|------|------|---|-----------------------------------------------|---------------------------|------------------|--|
| $\{\phi_{[\text{MAX}]}\}$ | V                                          |      | O    | D |                                               | $\{\phi_{[\text{MAX}]}\}$ | X                |  |
|                           | ukiriŋa <sup>H</sup>                       | ĩ    | ɿó   | — | y <sup>H</sup> ru <sup>0</sup> u <sup>H</sup> |                           | ó <sup>H</sup> . |  |
|                           | u-kiriŋ-aŋ                                 | i    | do   | — | guru-u                                        |                           | o                |  |
|                           | 1ERG-send-APPL                             | 3ABS | that |   | teacher-1GEN                                  |                           | there            |  |
|                           | 'I sent that (thing) there to my teacher.' |      |      |   |                                               |                           |                  |  |

This narrow result suggests that VI is not implicitly constrained by c-command: it can access and respond to the phonological shape of material that sits above the  $X^0$ s that it targets. In tandem with our earlier findings on the sensitivity of VI to high-level patterns of phonological phrasing, we thus arrive at a final line of evidence that VI can operate within a phonological evaluation that is global in character.

## 5 Conclusion

This investigation of Mandar has shown that certain  $X^0$ s are forced to supplete in a calculus that sees more phonological information—and is more sensitive to phonological constraints—than is typically allowed by standard models of VI in DM. The distribution of top-down suppletion shows that VI must target certain  $X^0$ s in awareness of their position in the  $\phi$ , the metrical structure of the left edge of the following  $\phi$ , and the organization of both  $\phi$ s into higher prosodic domains. The irrelevance of c-command and the consistent surface orientation of this system suggest that VI must target these  $X^0$ s in the global phonological evaluation where the surface prosody is built. The symmetry between suppletion and segmental repairs in the system of Positional Minimality, in turn, suggests that VI must be able to be guided by output-oriented phonological constraints. The result is a broader case that VI must operate in the phonology.

This conclusion opens up several shifts in perspective around the broader architecture of the postsyntax. On a modular and feed-forward theory of the grammar, to begin, phonological processes should neither feed nor drive operations in any prior module. As a result, VI should be unable to trigger any operation in the morphosyntax: not fission (Harbour 2008) and not movement (Starke 2009). It follows that such operations must occur for morphosyntactic reasons alone (and on fission, see Hewett 2022).

The second result lies with the theory of silence. The phonology now seems to be responsible for suppressing copies in the chains created by syntactic movement (Chomsky 1993), as phonological considerations often determine the identity of the copies selected for suppression (Bošković 2001). Bennett, Elfner, and McCloskey (2019) make a parallel case that the phonology must be responsible for silencing the constituents that are marked for ellipsis, as phonological constraints can determine the extent of suppression in ellipsis sites. On the hypothesis that the phonology is also the derivational locus of VI, it would then become possible to understand both chain reduction and ellipsis as cases of phonologically guided blocking of VI (Raffelsiefen 1996, 1999, 2004).



The third result lies with the cycle. By situating VI in the phonology, the framework developed here imposes no intrinsic limit on the amount of phonological context that is accessible to VI. This step provides a natural way to understand a set of parallel cases in which VI is conditioned by top-down context. Beyond the alternations that emerge in Mandarin and English when functional  $X^0$ s are stranded at the edge of the  $\phi$ , it is common for  $X^0$ s to supplete in response to their position in the  $\iota$  (Kenstowicz 2005, Henderson 2012, Royer 2022). In Mandarin, for instance, there is a  $v^0$  that takes the form  $-i$  at the right edge of the  $\iota$  (93a). When this  $v^0$  is not final in the  $\iota$ , it is not exponed (93b): either suppling to a null allomorph or being phonologically blocked from undergoing VI.

(93) *Top-down suppletion: The intonational phrase*

- a.  $\{_{\iota}$  á      mu<sup>H</sup>pájinoì<sup>H</sup>      }?  
       a      mu-pajino-i  
           what 2ERG-play-TRANSITIVE  
       ‘What are you playing?’
- b.  $\{_{\iota}$  á      mu<sup>H</sup>pájino\_\_\_<sup>H</sup>      díó<sup>H</sup> }?  
       a      mu-pajino-i      dio  
           what 2ERG-play-TRANSITIVE there  
       ‘What are you playing over there?’

In the same vein, the framework developed here allows VI to apply to multiple  $X^0$ s in parallel. This move opens up a natural analysis for the many patterns of allomorphy that cannot be handled by a strictly bottom-up and serial form of VI. The most obvious cases of this type involve outward-sensitive phonologically conditioned suppletion, where VI is influenced by the phonological features of a higher head  $X^0$  as it targets a lower head  $Y^0$  (Svenonius 2013, Deal and Wolf 2017, Rolle and Bickmore 2022). One example of this pattern comes from French, where Swiggers (1985) presents the following paradigm: the root  $\sqrt{\text{EGG}}$  is realized in its plural form as  $\emptyset$  after DP-internal  $X^0$ s that end in  $/z/$  and in all other contexts as  $\text{æf}$  (Swiggers also claims that speakers avoid pronouncing this root after *neuf* ‘nine’, where its appearance is plausibly blocked in the phonology by a double- $\text{æf}$  constraint).

(94) *Outward-sensitive phonologically conditioned suppletion*

- a. les  $/\emptyset/$ , des  $/\emptyset/$ , trois  $/\emptyset/$ , six  $/\emptyset/$ , dix  $/\emptyset/$   
       the eggs some eggs three eggs six eggs ten eggs
- b. quatre  $/\text{æf}/$ , cinq  $/\text{æf}/$ , sept  $/\text{æf}/$ , huit  $/\text{æf}/$   
       four    eggs five eggs seven eggs eight eggs  
       (French; Swiggers 1985)

These conclusions thus raise a network of questions on the interpretation of earlier work on suppletion in DM. The tendencies that were once taken to reveal cyclicity in VI may be amenable to reanalysis in terms of the standard machinery of postcyclic phonology: reference to prosodic structure (Selkirk 1984, Inkelas 1989) and constraints on output-output faithfulness (Benua 1997) and paradigm uniformity (Burzio 1994, Kenstowicz 1996, Steriade 1999, Albright 2002). Syntactic

locality conditions on suppletion, such as those noted by Embick (2010) and Merchant (2015), may be integrated into the growing literature on how purely parallel phonological evaluation may be sensitive to the fine-grained syntactic properties of its input (Cheng and Downing 2016). All cases of suppletion that are not output-optimizing, in turn, can be understood through the ranking of PRIORITY over output-optimizing constraints (Bonet, Lloret, and Mascaró 2007).

Against this backdrop, the status of the cycle rises to center stage. As phonological theory began to address the types of top-down interactions discussed here, it began to move past the cycle and shift toward global and parallel frameworks (Prince 1975, Dresher 1983). As the theory of suppletion begins to grapple with the same domain, it may arrive at the same result. As an alternative, some role for the cycle may persist: for instance, to capture phonologically conditioned alternations that are surface-opaque (Embick 2010, Kalin 2020). On a theory that espoused some type of cycle, such as one couched in Stratal OT (Bermúdez-Otero 1999, Kiparsky 2000), it would then become necessary to develop a framework that allowed for the top-down interactions above.

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